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MTH 101 — Past Questions & Materials

by Renance

Course: MTH 101

Institution: Federal University of Technology, Akure

Sources in this pack: 5

Includes past questions and lecture notes

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Paper 1 of 5 — Cyssa hub100LVL materials1st SemesterMTH 101MATHEMATICAL INDUCTION - MTH 101.pdf

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P1

MATHEMATICAL INDUCTION

The set of natural numbers $\mathbb{N}$, the set of whole numbers $\mathbb{W}$ and the set of positive integers have the special properties of being well-ordered.

Definition: Least Element
 Let S be a non-empty subset of $\mathbb{Z}$.
 An element $a \in S$ is called a least element (or minimum element) of S , if $a \leq b \forall a, b \in S$.

Example

- ① Natural number has a least element namely 1
- ② Set of positive integer has a least element 1, while set of integer has no least element.

Definition: Well-Ordering Principle
 Well-ordering Principle states that every non-empty subset of $\mathbb{N}$ has a least (or minimum) element.

The principle of mathematical induction applies to the sets of natural numbers, Whole numbers and integers because they are well-ordered.

PRINCIPLE OF MATHEMATICAL INDUCTION

Let $P(n)$ be a statement about a positive

integer n .

The principle of Mathematical induction

states that if:

- (i) $P(1)$ is true (i.e. $P(n)$ is true for $n=1$) and
- (ii) $P(k)$ is true for $\text{some } k \in \mathbb{N}$, implies

that it is true for $P(k+1)$.

Then, $P(n)$ is true for all $n \in \mathbb{N}$.

We also have the following theorem

Theorem:

Let $P(n)$ be a statement about a positive

integer n such that:

- (i) $P(1)$ is true (i.e. $P(n)$ is true for $n=1$)
- (ii) $P(m)$ is true for all positive integers

$m < k$, implies $P(k)$ is true.

Then, $P(n)$ is true for all $n \in \mathbb{N}$.

Example:

Prove that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

for every $n \in \mathbb{N}$

Proof

Let $S_n = 1 + 2 + 3 + \dots + n$ (LHS)

and $P(n)$ be a statement that

$$S_n = \frac{n(n+1)}{2} \quad \text{(RHS)}$$

for $n=1$.

$$\text{Thus, } S_1 = 1 \quad \text{(LHS)}$$

Also,

$$S_1 = 1(1+1) = 1 \quad (RHS)$$

Since, LHS = RHS
 $\therefore$ It is true for $n=1$
 Thus, $P(1)$ is true.

Now, assume, it is also true for $n=k$, we have,

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

For $n=k+1$, we obtain;

$$1 + 2 + 3 + \dots + k + (k+1) = \frac{k(k+1) + (k+1)}{2}$$

$$= \frac{k^2 + 3k + 2}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$= \frac{(k+1)(k+1+1)}{2}$$

$\Rightarrow$ It is true for $n=k+1$
 $\therefore$ If it is true for $n=k$, it is true for $n=k+1$
 Thus, $P(n)$ is true.
 Hence, by the principle of induction,
 $P(n)$ is true for all $n \in \mathbb{N}$

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P₄

Due

Show by induction that:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for every $n \in \mathbb{Z}$

Proof

Let $S_n = 1^2 + 2^2 + 3^2 + \dots + n^2$ (LHS) and $P(n)$ be a statement that

$$S_n = \frac{n(n+1)(2n+1)}{6} \quad \text{(RHS)}$$

For $n=1$

$$S_1 = 1^2 = 1 \quad \text{(LHS)}$$

Also, $S_1 = \frac{1(1+1)(2(1)+1)}{6} = \frac{(2)(3)}{6} = 1 \quad \text{(RHS)}$

Since, LHS = RHS = 1

$\Rightarrow$ It is true for $n=1$

Thus, $P(1)$ is true.

Now, assume it is true for $n=k$, we have

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

For $n=k+1$, we obtain

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

Exercise
Use mathematical induction to prove that

(1) $1^3 + 2^3 + \dots + n^3 = n^2(n+1)^2$ for every $n \in \mathbb{N}$

(2) $1 + 3 + \dots + (2n-1) = n^2$

(3) $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \frac{n(n+1)}{2}$

(4) $1 \cdot 2 \cdot 4 + 2 \cdot 3 \cdot 5 + \dots + n(n+1)(2n+1) = \frac{1}{12} n(n+1)(n+2)(8n+13)$

(5) $1^3 + 2^3 + 3^3 + \dots + (2n-1)^3 = n^2(2n^2-1)$

(6) $\sum_{k=1}^n \frac{1}{n(n+1)} = \frac{n+1}{n}$

PS (11)

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MTH 101 UPDATE

ATTENTION: ALL MTH 101 CARRY-OVER STUDENTS

MTH 101 assignment serves as Continuous Assessment (CA) for MTH 101 2024/25 Academic Session. All students who are carrying over MTH 101 are required to submit their assignments to the office of the coordinator, Dr. O.O. Ajijola located on the 1st floor of the Central Lab, opposite the Mathematical Sciences Department.

DATE: Thursday, 27/3/25 TIME: 8.30 AM - 1.00 PM

DATE: Friday, 28/3/25 TIME: 8.00 AM - 3.30 PM

Strictly note that any student who fails to submit the MTH 101 assignment will have no marks for the Continuous Assessment (CA).

Dr. O.O. Ajijola
MTH 101 Coordinator
2024/2025 Academic Session

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MTH101 CBT CA QUESTIONS (2024/25)

1. According to mathematical induction, which of these is a factor of $2^{n+1}-1$?
A. 3
B. 5
C. 8
D. 10
2. A school went to a competition with a contingent of 11 football players and 8 sprinters. If 3 sprinters are also footballers, how many students are either footballers or sprinters?
A. 16
B. 17
C. 18
D. 19
3. What is the sum to infinity of a geometric progression with first term 32 and common ratio $\frac{1}{2}$?
4. Insert five arithmetic means between x and $5x - 4$.
5. The common difference of an arithmetic progression (AP) is:
A. The difference between consecutive terms
B. The difference between the first and last term
6. $A \setminus (B \cup C)$ is:
A. $A - (B \cup C)$
B. $A \cap B \cap C$
C. $(A \cap B)'$
D. $A \cup B$
7. What is the sixth term of the sequence 1, 3, 9, ... ?
8. Insert five arithmetic means between 3 and 12
9. The sum of the first 5 terms of an A.P where $a = 2$ and $d = 3$ is:
A. 40
B. 25
C. 30
D. 35
10. If $a = 10$, $b = 20$, and $c = 30$ in an A.P, the arithmetic mean of a and b is:
A. 21
B. 15
C. 18
D. 9

11. Find the 8th term of a G.P whose first term is 3 and the common ratio is 2.
A. 384
B. 320
C. 364
D. 260
12. Determine the sum of the first four terms of the series $36 + 24 + 16 + \dots$
A. $265/3$
B. $65/3$
C. $250/3$
D. $260/3$
13. If A, B, and C are any three sets, what is $A \cap (B \cup C)$?
A. $(A \cap B) \cup (A \cap C)$
B. $(A \cup B) \cap (A \cup C)$
C. $(A \cup B) \cup (A \cup C)$
D. $(A \cap B) \cap (A \cap C)$
14. Given that $A = \emptyset$. Find the cardinality of $P(A)$.
A. No element
B. 0
C. 2
D. 1
15. Insert the arithmetic mean between 10 and 20.
16. Find the sum to infinity:
 $-2 + 2/3 - 2/9 + \dots$
A. -3
B. $3/2$
C. 3
D. $-3/2$
17. The third term of a geometric progression is 63, and the fifth term is 467. Find the sum of the first 4 terms.
18. Given the universal set $U = \{5, 6, 7, 8\}$, $P = \{6, 7\}$, and $Q = \{6, 8\}$, find $P \cap Q$.
19. $2^{kn} - 1$ is divisible by:
A. 3
B. 8
20. Find the cardinality of natural numbers less than 11.
21. Find the sum of the first five terms of the series: $8 + 4 + 2 + \dots$

22. For every positive integer n , $7^n - 3^n$ is divisible by:
- A. 3
 - B. 6
 - C. 4
 - D. 5
23. What is the cardinality of the power set $\{x : x \in \mathbb{N}, x < 11\}$?
- A. 1023
 - B. 2048
 - C. 1024
 - D. 10
24. Given the sequence $2, 2/5, 2/25, \dots$ what is the fifth term?
25. Insert the three arithmetic means between $x - 2$ and $x + 38$.
26. In the principle of induction, the propositional statement $P(n)$ is true for n in the set of:
- A. Rational numbers
 - B. Real integers
 - C. Natural numbers
 - D. Integers
27. In a class of 60, 40 students speak Hausa, 19 students speak Igbo, and 7 students speak both Hausa and Igbo. How many students speak neither Hausa nor Igbo?
- A. 8
 - B. 6
 - C. 7
 - D. 9
28. If α and β are the roots of the quadratic equation $2x^2 - 7x + 5 = 0$, find $\alpha\beta$ (ab).
- A. $5/1$
 - B. $-5/2$
 - C. $7/2$
 - D. $-7/5$
29. Given the sequence $-1/3, 1/2, \dots$ find the product of a and r .
30. For a quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, when $D < 0$, then the roots are:
- A) Real and equal
 - B) Real and distinct
 - C) Complex and distinct
 - D) Rational and distinct
31. If $P(A) = \emptyset$, find the cardinality of $P(A)$.
- A) 0

- B) 1
- C) 2
- D) Infinite

32. By the principle of mathematical induction, 2^{4n-1} is divisible by:

- A) 8
- B) 5
- C) 3
- D) 10

33. Find the sum of the first four terms of the series $1 + \sqrt{3} + 3 + \dots$

- A) $4(\sqrt{3} - 1)$
- B) $2(\sqrt{3} + 1)$
- C) $2(\sqrt{3} - 1)$
- D) $4(\sqrt{3} + 1)$

34. Let $A = (\text{empty set})$. Find $P(A)$.

- A) 0
- B) 1
- C) 2
- D) Infinite

35. Find the 6th term of the geometric progression: $1 + 3 + 9 + \dots$

- A) 243
- B) 729
- C) 81
- D) 162

36. Given the quadratic equation $2x^2 - 7x + 5$, find $a^3 + b^3$.

- A) 125
- B) 343
- C) 216
- D) 512

37. What is the formula for the sum of terms in an arithmetic progression (AP)?

- A) $S_n = n/2 (2a + (n-1)d)$
- B) $S_n = n(a + l)/2$
- C) $S_n = a(1 - r^n) / (1 - r)$
- D) $S_n = ar^n - 1$

38. In the principle of mathematical induction, the propositional statement $P(n)$ is true for n in the set of:

- A) Real numbers
- B) Natural numbers
- C) Integer numbers
- D) Rational numbers

39. Find the 4th term of a geometric progression whose first term is 6 and common ratio is 3.
A) 486
B) 192
C) 120
D) 162
40. Find the sum of the first 6 terms of an arithmetic progression where $a = 2$ and the last term $l = 20$.
A) 66
B) 54
C) 84
D) 72
41. Find the sum of the first 5 terms of a sequence whose first and last terms are 3 and 3 respectively.
A) 360
B) 243
C) 242
D) 363
42. The sum of the first n terms of an arithmetic progression is given by $S_n = 5n^2 - 3n$. Find the 10th term.
A) 92
B) 96
C) 80
D) 94
43. Insert three geometric means between x/p^2 and xp^2 .
A) $1/p, x, p$
B) $x/p, p, xp$
C) $x/p, x, p$
D) $x/p, x, xp$
44. Determine the nature of the roots of the equation $x^2 - 2x + 1 = 0$
A. Real and equal
B. Real and distinct
C. Imaginary
D. None of the above
45. If the quadratic equation $2x^2 - 7x + 5 = 0$ has α and β as its roots, find $\alpha^2 + \beta^2$
A. 17
B. 25
C. 29
D. 35
46. What will be $P(k+1)$ for $P(n) = n^3(n+1)$?
A. $(k+1)^3(k+2)$

B. $(k+1)^3(k+1)$

C. $k^3(k+2)$

D. $k^3(k+1)$

47. Insert three arithmetic means between $x + 7y$ and $5x + 3y$

A. $2x + 6y, 3x + 5y, 4x + 4y$

B. $2x + 8y, 3x + 6y, 4x + 5y$

C. $2x + 7y, 3x + 5y, 4x + 3y$

D. $2x + 6y, 3x + 4y, 4x + 2y$

48. What are the five numbers between 3 and 12 in an arithmetic progression?

A. 5, 7, 9, 11, 13

B. 5, 7, 9, 11, 12

C. 4, 6, 8, 10, 12

D. 4, 5, 7, 10, 12

49. For $P(n)$, n is true for n in the set of:

A. Natural numbers

B. Integers

C. Real numbers

D. Whole numbers

50. Insert five arithmetic means between 3 and 12

A. 5, 6.8, 8.6, 10.4, 12

B. 4, 5.8, 7.6, 9.4, 11.2

C. 4, 6, 8, 10, 12

D. 3.4, 4.8, 6.2, 7.6, 9

51. What is the nature of the roots of the equation $2x^2 - 3x + 5 = 0$?

A. Real and equal

B. Real and distinct

C. Imaginary

D. Cannot be determined

52. For the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, where $D > 0$, what is the nature of the roots?

A. Real and equal

B. Real and distinct

C. Imaginary

D. None of the above

53. $P = \{x : x \leq 3\}$, $Q = \{x : 5 < x < 12\}$, $R = \{x : -2 \leq x < 5\}$. What is $P \cap (Q \cup R)$?

A. $-4 \leq x \leq 3$

B. $x < 0$

C. $x < -4$

D. $\emptyset$

54. What is the sum of the first four terms of the series $1 + \sqrt{3} + 3 + \dots$?

- A. $2(\sqrt{3} + 1)$
- B. $4(\sqrt{3} + 1)$
- C. $4(\sqrt{3} - 1)$
- D. $2(\sqrt{3} - 1)$

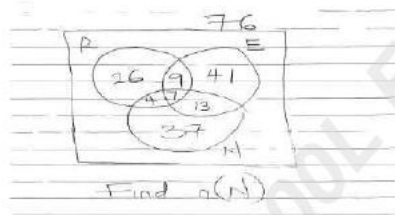
55. The first term of an arithmetic progression is 2, the last term is 50, and the common difference is 4. What is the number of terms in the AP?

- A. 10
- B. 12
- C. 13
- D. 15

56. The mean of two terms is 18. If the first term is 12, what is the second term?

- A. 18
- B. 24
- C. 26
- D. 30

56.



57. In an A.P, the sum of the first n terms is $S_n = 2n^2 + 3n$. What is the n th term (a_n)?

- A. $4n + 1$
- B. $4n - 1$
- C. $2n + 3$
- D. $3n^2$

58. What will be $P(k+1)$ for $P(n) = n^3(n + 1)$?

- A. $(k + 1)^3(k + 2)$
- B. $k^3(k + 1)$
- C. $(k + 1)^3(k + 1)$
- D. $k^3(k + 2)$

59. Find the sum to infinity of $32 + 16 + 8 + \dots$

- A. 64
- B. 60
- C. 20
- D. 128

60. Insert three arithmetic means between $x + 7y$ and $5x + 3y$

- A. $2x + 6y, 3x + 5y, 4x + 4y$
- B. $2x + 8y, 3x + 6y, 4x + 5y$
- C. $2x + 7y, 3x + 5y, 4x + 3y$
- D. $2x + 6y, 3x + 4y, 4x + 2y$

61. Find the sum to infinity of $-2 + \frac{2}{3} - \frac{2}{9} + \dots$

- A. -2
- B. -3
- C. $-\frac{6}{7}$
- D. $-\frac{3}{2}$

62. Determine the sum of the first five terms of the series $1 + 3 + 9 + \dots$

- A. 121
- B. 221
- C. 242
- D. 321

63. Find the 5th term of the sequence $2, \frac{2}{5}, \frac{2}{25}, \dots$

- A. $\frac{2}{125}$
- B. $\frac{2}{3125}$
- C. $\frac{2}{625}$
- D. $\frac{2}{15}$

64. If $a = 10, b = 20, c = 30$, what is the arithmetic mean of a and b ?

- A. 15
- B. 16
- C. 19
- D. 20

65. In a class of 60 students, 40 speak Hausa, 19 speak Igbo, and 7 speak both. How many speak neither?

- A. 11
- B. 6
- C. 8
- D. 7

66. By the principle of mathematical induction, $2^n(4n - 1)$ is divisible by

- A. 10
- B. 5
- C. 8
- D. 3

67. Insert a geometric mean between $x(x + 1)/(x - 2)$ and $x((x + 1)^3)/(x - 2)^3$

68. The 2nd term of a G.P is 9, and the 4th term is 81. Find the first term.

- A. 3
- B. 9
- C. 2
- D. 6

69. Find the sum of the first six terms of an A.P where the first term $a = 2$ and the last term $l = 20$.

- A. 84
- B. 66
- C. 72
- D. 80

70. For a quadratic equation $ax^2 + bx + c = 0$, if the determinant is less than zero, describe its roots.

- A. Real and distinct
- B. Real and repeated
- C. Not real roots

71. Insert four arithmetic means between 243 and -1

- A. 81, 27, 9, 3
- B. 128, 64, 32, 16
- C. 100, 75, 50, 25
- D. 90, 60, 30, 15

72. What is the cardinality of the power set $\{x : x \in \mathbb{N}, x < 11\}$?

- A. 1024
- B. 1023
- C. 10
- D. 11

73 - 75.

1) Insert 4 geometric mean between $a(2+1)$ and $a(2+1)^4$

A. $\frac{(2+1)^2}{(2-2)^2}$ B. $\frac{2(2+1)^2}{(2-2)^2}$

C. $\frac{2(2+1)}{(2-2)^2}$ D. $\frac{(2+1)^2}{(2-2)^2}$

2) If $P(n) = n^2(n+2)(n^2-1)$ is false, then which is $P(k+1)$?

A. $k(k+1)^2(k+2)(k+1)$

B. $(k+1)^2(k+1)(k^2-1)$

C. $(k+1)^2(k+2)(k^2-1)$

D. $(k+1)(k+2)(k+1)$

3) Given that the universal set U has subsets $P = \{x: x < 3\}$ & $Q = \{x: 1 < x < 4\}$

Find $P \cap Q$

A. $-1 < x < 3$

B. $-1 < x < 4$

C. $-1 < x < 3$

D. $-1 < x < 4$

76. In an A.P, the sum of the first n terms is $S_n = 2n^2 + 3n$. What is the n -th term (a_n)?

- a. $4n - 1$

- b. $4n + 1$
- c. $3n - 1$
- d. $3n + 1$

77. What will be $P(k+1)$ for $P(n) = n^3(n+1)$?

- a. $k(k+1)(k+2)(k+4)$
- b. $(k+1)^2(k+4)(k^2 - 2)$
- c. $(k+1)(k+2)(k+4)$
- d. $(k+1)^2(k+3)(k^2 - 1)$

78. Find the sum to infinity of $32 + 16 + 8 + \dots$

- a. 64
- b. 60
- c. 20
- d. 128

79. Insert three arithmetic means between $x + 7y$ and $5x + 3y$.

- a. $2x, 2x+2, 3x+3$
- b. $2x+1, 3x+2, 4x+3$
- c. $2x+1, 3x+2, 4x+4$
- d. $x+2, 2x+3, 3x+4$

80. Find the sum to infinity of $-2 + 2/3 - 2/9 + \dots$

- a. -3
- b. -2
- c. -4
- d. -1

81. Determine the sum of the first five terms of $1 + 3 + 9 + \dots$

- a. 221
- b. 242
- c. 321
- d. 121

82. Find the 5th term of the sequence $2, 2/5, 2/25, \dots$

- a. $2/625$
- b. $2/125$
- c. $1/125$
- d. $1/625$

83. If $a = 10$, $b = 20$, $c = 30$, find the arithmetic mean of a and b .

- a. 19
- b. 16
- c. 20
- d. 15

84. In a class of 60 students, 40 speak Hausa, 19 speak Igbo, and 7 speak both. How many speak neither?
- 11
 - 6
 - 8
 - 7
85. By the principle of mathematical induction, 2^{4n-1} is divisible by:
- 3
 - 5
 - 10
 - 8
86. The 2nd term of a G.P is 35, and the 4th term is 875. Find the 3rd term.
87. Insert three geometric means between x/p^2 and xp^2 .
88. If the 8th term of a G.P is 4374 and the 5th term is 162, find the sum of the first four terms.
89. For a quadratic equation $ax^2 + bx + c = 0$, if the determinant is less than zero, describe its roots.
- Real and repeated
 - Not real roots
 - Real and distinct
90. Insert four arithmetic means between 243 and -1.
- 81, 27, 9, 3
 - 128, 64, 32, 16
 - 128, 64, -32, 16
 - 81, 27, -9, 3
91. Find the 8th term of a G.P whose first term is 3 and the common ratio is 2.
- 384
 - 260
 - 320
 - 364
92. In an A.P, the first term is 2, the last term is 50, and the common difference is 4. How many terms are there in the series?
- 12
 - 13
 - 14
 - 15
93. Find the sum of the first six terms of an A.P where the first term $a = 2$ and the last term $l = 20$.
- 84
 - 66
 - 72

d. 78

94. What is the cardinality of the power set $\{x : x \in \mathbb{N}, x < 11\}$?

- a. 1024
- b. 2048
- c. 1023
- d. 10

95. The sum of n terms is given as $S_n = 2n^2 - 3n$. Find the n -th term.

- a. $4n - 1$
- b. $4n + 1$
- c. $3n - 1$
- d. $3n + 1$

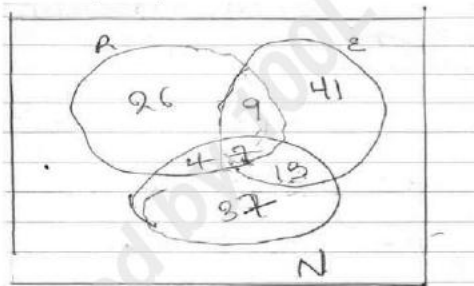
96. For what value of n , n^2 is true?

- a. $n \geq 5$
- b. $n \geq 2$
- c. $n \geq 1$
- d. $n \geq 4$

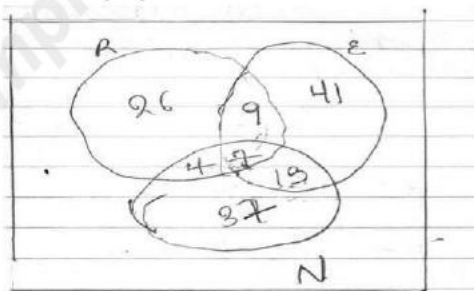
97. Find the number that can divide $7^n - 3^n$ when n is a positive integer.

98. If $n(P \cup Q) = 50$, $n(P) = 40$, and $n(Q) = 18$, find $n(P \cap Q)$.

99. Find $n(R \cap E \cap N)$



100. Find $n[E - R]$



9. Insert three arithmetic mean between x and $5x + 4$

10. Given $x^2 + 2x + 1 = 0$, what is the value of $\alpha^3 + \beta^3$ if α and β are the roots of the equation.
(A) 8 (B) 4 (C) 1 (D) 2

11. The first term of a G.P is 81 and fourth term is. Find sum to infinity
(A) $85\frac{1}{2}$ (B) $47\frac{1}{3}$ (C) 43 (D) $121\frac{1}{2}$

1. What is $P(k+1)$ in $P(n) = n^2(n+3)(n^2-1)$

2. $P(n) = n^3(n+1)$ forgot what we were to do

3. 2^{4n-1} what can divide it a) 3, b) 5 c) 8

4. Insert one geometric mean between

$$\frac{x(x+1)}{x-2}, \frac{x(x+1)^3}{(x-2)^3}$$

$$3) P = \{x: x \leq 3\}$$

$$Q = \{x: 5 < x \leq 12\}$$

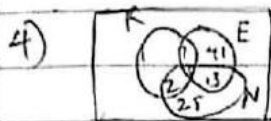
$$R = \{x: 3 < x \leq 4\}$$

$$\text{Find } P \cap (Q \cup R)^c$$

$$A \{x: x \leq -5\}$$

$$B \emptyset$$

$$C$$



Not sure of the numbers

$$U = \{1, 2, 7, 9, 13, 25, 41\}$$

and R, P and E are subsets of U

$$\text{Find } n(E - R)$$

$$A 7 \quad B 2 \quad C \{13, 41\} \quad D \{7, 41\}$$

$$5) \text{ Find } n(R \cap E \cap N)$$

$$A 7 \quad B 1 \quad C$$

$$6) \text{ Find the sum of the first 6}$$

terms of a series $18, 1, 3, \dots$

$$A 728 \quad B 729 \quad C 27 \quad D 27$$

$$\frac{27}{27} \quad \frac{27}{27} \quad \frac{728}{728} \quad \frac{729}{729}$$

$$190 + \frac{726}{27}$$

$$10) -2, \frac{2}{3}, -\frac{2}{3}$$

Find the sum to infinity

$$1) \text{ Insert 2 arithmetic mean between } (x-2) \text{ and } (x+38)$$

$$(x+8), (x+18), (x+28)$$

$$2) \text{ The sum of the } n^{\text{th}} \text{ term of an A.P is } 5n^2 - 3n$$

Find the 10th term

$$A 92 \quad B 94 \quad C 96 \quad D$$

$$7) \text{ The nature of the root of an equation } x^2 - 22x + 1 \text{ is}$$

$$A \text{ Repeated and imaginary}$$

$$B \text{ Repeated and real}$$

$$C \text{ Real and distinct}$$

$$8) \text{ For set } A, B, \text{ and } C$$

$$A \cap (B \cup C) \text{ is}$$

$$A (A \cap B) \cap (A \cap C)$$

$$B (A \cup B) \cup (A \cup C)$$

$$C (A \cap B) \cup (A \cap C)$$

$$9) \text{ If } P = \{x: -4 \leq x < 13\}$$

$$Q = \{x: -2 < x < 5\}$$

$$\text{Find } (P \cup Q)$$

$$A P \quad B Q \quad C \{x: -2 < x < 5\}$$

$$D \emptyset$$

Suppose $x \neq -1$ for what values of x will

$x, \frac{x(x-2)}{(x+1)}, \frac{x(x-2)^2}{(x+1)^2}$ has a sum to infinity

a. $\left(\frac{x-2}{x+1}\right) \leq 1$

b. $x < -1$

c. $\frac{x-2}{x+1} < 1$

d. $\frac{x(x-2)}{x+1} < 1$

Find $\alpha\beta$.

$$2x^2 - 7x + 5 = 0$$

has α & β as its roots

Find $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$.

$$D > 0$$

$$a^2 + b^2 + c^2 \quad \text{the } D \neq 0$$

1. What is $P(n+1)$ in $P(n) = n^2(n+3)(n^2-1)$

2. $P(n) = n^3(n+1)$ forgot what we were to do

3. 2^{4n-1} what can divide it a) 3, b) 5 c) 8

4. Insert one geometric mean between

$$\frac{x(x+1)}{x-2}, \frac{x(x+1)^3}{(x-2)^3}$$

5. $36, +24 + 16 \dots \dots$ find the sum

6. $x^2 - 5x + 6,$

- a) the root is undefined
- b) no real root
- c) real and equal root
- d) real and distinct root

In a quadratic equation $ax^2 + bx + c$ where the root is given by $D > 0$. $a \neq 0$

- a) The root is repeated and no real root
- b) Undefined
- c) Real and equal
- d) no real root

In a quadratic equation $ax^2 + bx + c = 0$ where the root is given by $D \gg 0$. $a \neq 0$.

- a) The root is repeated and no real root
- b) Undefined
- c) Real and equal
- d) no real root

5 $36, +24 + 16 \dots \dots$ find the sum

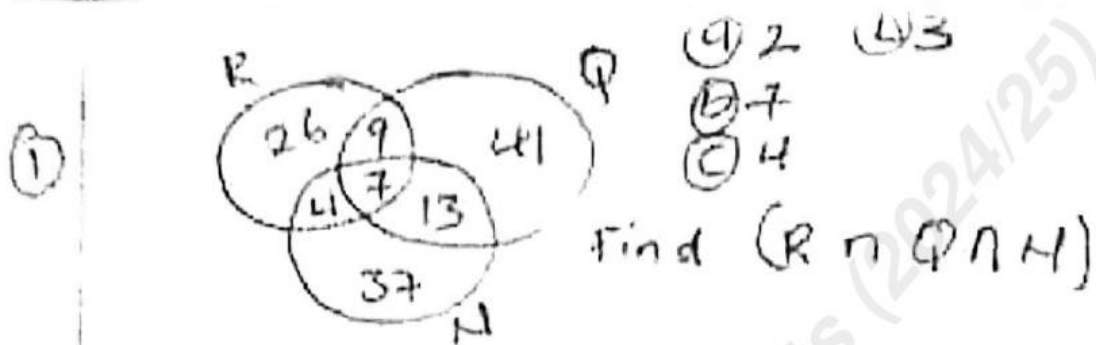
6 $x^2 - 5x + 6,$

a) the root is undefined

b) no real root

c) real and equal root

d) real and distinct root



② For every positive integer "n" $7^n - 3^n$ is divisible by ?

- (A) 5 (B) 3 (C) 6 (D) 4

③ Find the sum of infinity of $32 + 16 + 8 + \dots$

- (A) 64 (B) 60 (C) 128 (D) 82

④ Insert Three A.M b/w $x + 7y$ and $5x + 3y$.

- (A) $2x + 6y, 3x + 5y, 4x + 4y$
 (B) $2x + 5y, 3x + 4y, 4x + 3y$
 (C) $x + 6y, 3x + 5y, x + 4y$
 (D) $2x + 6y, 3x + 4y, 4x + 2y$

The question I was able to Recall
 $2x^2 - 7x + 5$ has α & β
as the roots.
Find $\alpha\beta$.

Find ~~the~~
 ① The Power Set of A in
 $A = \emptyset$

Ⓐ 1

Ⓑ 2

Ⓒ 0

Ⓓ empty set

② Insert 3 G.M between
 x^2 and x^2

P

Ⓐ $\frac{x}{P}, x, P$

P

Ⓑ $\frac{1}{P}, x, xP$

P

Ⓒ

Ⓓ

③ if $P = 40$ and $Q = 18$

$$n(P \cup Q) = 50$$

Find

$$n(P \cap Q)$$

Ⓐ 108

Ⓓ

Ⓑ 58

Ⓒ 32

5th term of the sequence

$$2, \frac{2}{3}, \frac{2}{25}$$

☐ $\frac{2}{335}$
 ☐ $\frac{2}{225}$
 ☐ $\frac{2}{265}$
 ☒ $\frac{2}{435}$

If The 5th term of GP is 162 and 8th is 4374 find the product of a and r

☒ 6
 ☐ 3
 ☐ 12
 ☐ 9

The ~~6th~~^{6th} term of an AP is 20

10th term is 32 Find the

sum of the first 10 terms

☐ 185
 ☐ 200
 ☐ 195
 ☐ 225

Find the sum of infinity of
 $-2 + \frac{2}{3} - \frac{2}{9} + \dots$

* Given that $A = \emptyset$. Find the Cardinality of $P(A)$

- (a) 0
- (b) 2
- (c) 1
- (d) No element

* For every positive integer n , $7^n - 3^n$ is divisible by...

- (a) 5
- (b) 6
- (c) 3
- (d) 4

What is the Sum of Infinity of the geometric Series

$$x - x^2 + x^3 - x^4 + \dots$$

* If $U = \{5, 6, 7, 8\}$, $P = \{6, 7\}$ and $Q = \{6, 8\}$.
Find $(P \cap Q)$

To find $2^{2n} - 1$

Let $P(n)$ be the statement that $2^{2n} - 1$ is divisible by 3 for $n = 1$;

Hence, $2^{2(1)} - 1 = 3$, which is divisible by 3; thus $P(1)$ is true.

If $P(k)$ is true for some Integer, $k \geq 1$

i.e. $2^{2k} - 1$ is divisible by 3

Hence,

$$2^{2k} - 1 = 3m$$

Inductive step

If $P(k+1)$ is true;

$2^{2(k+1)} - 1$ is divisible by 3

$$2^{2(k+1)} - 1 = 2^{2k+2} - 1$$

Applying Indices

$$2^{2k} \times 2^2 - 1$$

$$= 4 \cdot 2^{2k} - 1$$

$$= \boxed{4 \cdot 2^{2k}}$$

$$4 \cdot 2^{2k} + 3 - 4$$

$$= \boxed{4(2^{2k})} (1)(2^{2k} - 4) + 3$$

$$4(2^{2k} - 1) + 3$$

$$\text{but } 2^{2k} - 1 = 3m$$

$$4(3m) + 3$$

$$= 12m + 3$$

Factor out 3

$$3(4m + 1)$$

hence, $P(k+1)$ is divisible by 3

Hence, 3 is a factor of $2^{2n} - 1$ (A).

$$2. \quad n(F) = 11$$

$$n(S) = 8$$

$$n(F \cap S) = 3$$

$$\text{Hence, } n(U) = n(F) + n(S)$$

$$n(U) = 11 + 8$$

$$= 19 \text{ sportsmen}$$

(D)

$$\cancel{n(F')} = 11 - 3$$

$$= 8$$

$$n(S') = 8 - 3$$

$$= 5$$

$$3. \quad a = 32$$

$$r = \frac{1}{2}$$

$$S_{\infty} = \frac{a}{1-r^n}$$

$$= \frac{32}{1-\frac{1}{2}}$$

$$= \frac{32}{\frac{1}{2}}$$

$$= 64$$

$$\frac{1}{2}$$

$$S_{\infty} = 64$$

$$1. \quad x: a, b, c, d, e, 5x-4$$

Hence,

$$a = x, T_1 = x$$

$$T_7 = a + 6d$$

$$5x-4 = x + 6d$$

$$5x-4-x = 6d$$

$$d = \frac{4x-4}{6}$$

$$d = \frac{4(x-1)}{6}$$

$$d = \frac{2}{3}(x-1)$$

Therefore

$$a = T_2 = a + d$$

$$= x + \frac{2}{3}(x-1)$$

$$= \frac{2x-2+x}{3}$$

$$= \frac{2x-2+3x}{3}$$

$$\frac{5x-2}{3}$$

$$b = T_3 = a + 2d$$

$$= x + \frac{4}{3}(x-1)$$

$$b = \frac{4x-4+3x}{3}$$

$$b = \frac{7x-4}{3}$$

$$c = T_4 = a + 3d$$

$$= x + \frac{6}{3}(x-1)$$

$$= x + 2(x-1)$$

$$= 3x-2$$

$$d = T_5 = a + 4d$$

$$= x + \frac{8}{3}(x-1)$$

$$= \frac{8x+3x-8}{3}$$

$$= \frac{11x-8}{3}$$

$$e = T_6 = a + 5d$$

$$= x + \frac{10}{3}(x-1)$$

$$= \frac{3x+10x-10}{3}$$

$$= \frac{13x-10}{3}$$

hence, the arithmetic means

$$\frac{5x-2}{3}; \frac{7x-4}{3}; \frac{3x-2}{3}; \frac{11x-8}{3}$$

$$\frac{13x-10}{3}$$

5. A.

6. $A \cap (B \cup C)$.

~~The set~~

Let's assume A to be a set of number such that $x \in x$ is even

$$A = \{2, 4, 6, 8, 10\}$$

Let B be a set of odd numbers

$$B = \{1, 3, 5, 7, 9\}$$

and lastly C be a consecutive

number from 1 to 5

$$C = \{1, 2, 3, 4, 5\}$$

Hence

$$n(C') = \{6, 7, 8, 9, 10\}$$

$$n(E) = \{1, 2, 3, 4, 5, 6, 7,$$

$$8, 9, 10\}$$

$$n(B \cup C') = \{1, 3, 5, 6, 7, 8, 9, 10\}$$

$$A \cap (B \cup C') = \{6, 8, 10\}$$

$$\text{Hence, } A \cap (B \cup C') = A$$

$$7 \quad 1^{\text{st}} \text{ no.} = 1$$

$$2^{\text{nd}} \text{ no.} = 3$$

$$\text{Common Ratio} = 3$$

Hence

$$T_n = ar^{n-1}$$

$$= 1 \times (3)^5$$

$$= 1 \times 729$$

$$= 729$$

$$8. \quad 3, a, b, c, d, e, 12$$

1st term = 3

$$\text{last term } (l) = 12$$

$$\text{Hence, } 12 = a + 6d$$

$$3 + 6d = 12$$

$$6d = 12 - 3$$

$$d = \frac{9}{6}$$

$$d = \frac{3}{2}$$

Hence,

$$a = a + d$$

$$= 3 + \frac{3}{2}$$

$$= 4.5$$

$$b = a + 2d$$

$$= 3 + 2\left(\frac{3}{2}\right)$$

$$= 3 + 3$$

$$= 6$$

$$c = a + 3d$$

$$= 3 + 3\left(\frac{3}{2}\right)$$

$$= 3 + \frac{9}{2} = 7.5$$

$$d = a + 4d$$

$$d = 3 + 4\left(\frac{3}{2}\right)$$

$$= 3 + 6$$

$$= 9$$

$$e = a + 5d$$

$$e = 3 + 5\left(\frac{3}{2}\right)$$

$$e = 3 + \frac{15}{2}$$

$$= 10.5$$

Hence, the Arithmetic means are;

4.5, 6, 7.5, 9, 10.5

$$9. \quad \sum_n = \frac{n(2a + (n-1)d)}{2}$$

$$\sum_5 = \frac{5(2(2) + 4(3))}{2}$$

$$= \frac{5(16)}{2}$$

$$= 5 \times 8$$

$$= 40 \quad (A)$$

10. $a = 10$, $b = 20$, $c = 30$

arithmetic mean of ab

$$= \frac{a+b}{2}$$

$$= \frac{10+20}{2}$$

$$= \frac{30}{2} = 15 \quad (B)$$

$$11. \quad T_n = ar^{n-1}$$

$$T_8 = ar^{8-1}$$

$$= ar^7$$

$$a = 3, r = 2$$

$$3 \times (2)^7$$

$$= 3 \times 128$$

$$= 384 \quad (P)$$

$$=$$

$$12. \quad a = 36$$

$$r = \frac{2}{3}$$

$$\text{Hence, } T_4 = ar^3$$

$$= 36 \times \left(\frac{2}{3}\right)^3$$

$$= 36 \times \frac{8}{27}$$

$$= 10.6$$

$$= \frac{32}{3}$$

$$\text{Since } r = \frac{2}{3} \text{ ; less than}$$

1

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$= \frac{36 \left(1 - \left(\frac{2}{3}\right)^4\right)}{1 - \frac{2}{3}}$$

$$= \frac{36 \left(1 - \frac{16}{81}\right)}{\frac{1}{3}}$$

$$= \frac{36 \left(1 - \frac{16}{81}\right)}{\frac{1}{3}}$$

$$= \frac{36 \left(\frac{65}{81}\right)}{\frac{1}{3}}$$

$$= \frac{2340}{81} \times \frac{81}{65}$$

$$= \frac{2340}{81} \times \frac{81}{65}$$

$$= \frac{36 \left(1 - \frac{16}{81}\right)}{1 - \frac{2}{3}}$$

$$= \frac{36 \left(\frac{65}{81}\right)}{\frac{1}{3}}$$

$$= \frac{2340}{81} \times \frac{81}{65}$$

Error

260

3

(8)

13. $A \cap (B \cup C)$

Assume, A

$A = \{2, 4, 6, 8, 10\}$

$B = \{1, 3, 5, 7, 9\}$

$C = \{1, 2, 3, 4, 5\}$

Hence,

$B \cup C = \{1, 2, 3, 4, 5, 7, 9\}$

$A \cap (B \cup C) = \{2, 4, 5\}$

But

$A \cap B = \emptyset$

Hence, $A \cap C = \{2, 4\}$

Therefore,

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$(A \cap B) \cup (A \cap C)$

$= (A)$

14. $A = \emptyset; P(A)$

No element (A)

15. $10, a, b, c, 20$

$a = 10$

$75 = a + 4d$

$20 = 10 + 4d$

$d = 10/4$

$a = a + d$

$= 10 + \frac{10}{4}$

$= 12.5 //$

$b = a + 2d$

$= 10 + \frac{20}{4}$

$= 10 + 5$

$= 15 //$

$c = a + 3d$

$= 10 + \frac{30}{4}$

$= 17.5 //$

16. Sum to Infinity

$$-2, \frac{2}{3}, -\frac{2}{9}$$

$$a = -2$$

$$r = -\frac{1}{3}$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{-2}{1 - (-\frac{1}{3})}$$

$$= \frac{-2}{1 + \frac{1}{3}} = \frac{-2}{\frac{4}{3}}$$

$$= -2 \times \frac{3}{4}$$

$$= -\frac{3}{2} \quad \textcircled{B}$$

17. $T_3 = ar^2 = 63$

$$T_8 = ar^7 = 467$$

$$S_4 = ?$$

If T_3 and T_8 are solved simultaneously,

$$ar^2 = 63 \quad \textcircled{1}$$

$$ar^7 = 467 \quad \textcircled{2}$$

Hence divide $\textcircled{2}$ by $\textcircled{1}$

$$r^5 = \frac{467}{63} \quad \frac{467}{63}a = 63$$

$$r = \sqrt[5]{\frac{467}{63}} \quad a = 63/\frac{467}{63}$$

$$r = 2.7 \quad a = 8.5$$

Hence, Sum of the first 4 terms

$$S_n = \frac{a(r^n - 1)}{r - 1} ; a = 8.5$$

$$S_n = 8.5 \left[\left(\frac{467}{63} \right)^4 - 1 \right]$$

$$2.7 - 1$$

$$S_{4th} = \frac{443.22}{1.7}$$

$$S_{4th} = \underline{260.7}$$

18. $U = \{5, 6, 7, 8\}$

$$P = \{6, 7\}$$

$$Q = \{6, 8\} ; Q' = \{5, 7\}$$

$$P \cap Q = \{6\} ; P \cap Q' = \{7\}$$

19. $2^{kn+1} - 1$ is divisible by 3, where k is an even integer, and n is a positive integer.

(A)

20. Cardinality of members less than 11
1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Hence, counting them there are 10 members.

21. $a = 8$

$$r = \frac{1}{2}$$

hence, sum of first 5

$$= \frac{a(1-r^n)}{1-r}$$

$$= \frac{8(1-(\frac{1}{2})^5)}{1-\frac{1}{2}}$$

$$= \frac{8(0.96875)}{0.5}$$

$$= \frac{7.75}{0.5}$$

$$= 15.5$$

$$22. 7^n - 3^n$$

Assume $n = 1$

$$7 - 3 = 4$$

thus 4 is divisible by 4

Assume $n = 2$

$$49 - 9$$

$$= 40$$

then 40 is divisible by 4

Hence, $7^n - 3^n$ is divisible by 4 (E)

23. The set $\{x: x \in \mathbb{N}, x < 11\}$

are natural numbers less than 11

which are $\{0, 1, 2, 3, 4, 5, 6, 7, 8,$

$9, 10\}$ This set has 11 elements

and a power set of n elements

is given as 2^n

Therefore the Cardinality
of a power set of
 $\{x: x \in U; x \subset U\} = 2^U$
and

$$2^U = 2048$$

(B)

24. $2, 2/5, 2/25, \dots$

the first = 2

Common ratio = $1/5$

$$T_5 = ar^4$$

$$= 2\left(\frac{1}{5}\right)^4$$

$$= 2 \times \frac{1}{625}$$

$$= \frac{2}{625}$$

25. $x-2, a, b, c, x+38$

$$a = x-2$$

$$T_5 = a + 4d$$

$$x+38 = x-2 + 4d$$

$$x+38 - x-2 = 4d$$

$$40 = 4d$$

$$d = \frac{40}{4}$$

$$d = 10$$

$$a = a + d$$

$$= x-2+10$$

$$= x+8$$

$$b = a + 2d$$

$$= x-2+20$$

$$= x+18$$

$$c = a + 3d$$

$$= x-2+30$$

$$= x+28$$

Hence, the arithmetic means are

$$x+8, x+18, x+28.$$

26. B

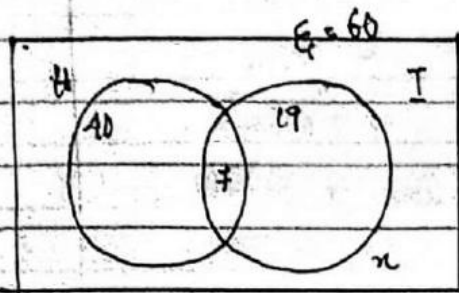
27. $n\left(\frac{1}{4}\right) = 60$

$$n(H) = 40$$

$$n(I) = 19$$

$$n(H \cap I) = 7$$

$$n(H \cap I) = ?$$



Hence, the no. of students that speak neither Hausa nor Igbo will be
 $n(E) = n(H) + n(I) + n(H \cap I) + n(\text{outside})$
 $66 = 40 + 19 + 7 + x$

$$66 = 66 - x$$

$$x = 66 - 60$$

$$x = 6 \text{ students}$$

(B)

28. α and β are roots of the

$$\text{eqn. } 2x^2 - 7x + 5 = 0$$

find $\alpha\beta$

$$\alpha\beta = \frac{c}{a}$$

$$\text{where } c = 5, a = 2$$

$$\alpha\beta = \frac{5}{2}$$

$$29. -\frac{1}{3}, -\frac{1}{2}i$$

$$a = -\frac{1}{3}$$

$$r = -\frac{1}{3} \times \frac{1}{2} = -\frac{1}{6}$$

$$i = -\frac{1}{3} \times \frac{1}{2} = -\frac{1}{6}$$

$$r = -\frac{3}{2}$$

hence, product of a and r

$$= -\frac{1}{3} \times -\frac{3}{2}$$

$$= \frac{1}{2}$$

$$30. ax^2 + bx + c = 0$$

where $a \neq 0$

$$D = b^2 - 4ac$$

when $D < 0$, the quadratic eqn. has no real roots.

Hence, (C)

31. $P(A) = \emptyset$

The cardinality of $P(A)$

is 2^0

$= 1$ (8)

32. $2^{(4n-1)}$

at $n=1$

$2^{(4-1)} = 2^3 = 8$

and is divisible by 8

at $P(k)$

$= 2^{4k-1} = 8m$

at $P(k+1)$

$= 2^{4(k+1)-1} = 8m$

$= 2^{4k+4-1}$

$$= 2^{4k+3}$$

$$= 2^{4k} \cdot 2^3$$

$$= 2^{4k} \cdot 8$$

$= 2^{4k-1+4}$

$= 2^{4k-1} \cdot 2^4$

$= 2^{4k-1} \cdot 16$

$= 16(2^{4k-1})$

but $2^{4k-1} = 8m$

$= 16 \times 8m$

$= 128m$

$= 8(16m)$

hence it is divisible by 8

(P)

33. $a=1$; $r=\sqrt{3}$

hence, S_n at $n=4$

$S_n = \frac{a(1-r^n)}{1-r}$ or $\frac{a(r^n-1)}{r-1}$

at $n=4$:

$$= \frac{1(1-(\sqrt{3})^4)}{1-\sqrt{3}}$$

$$= \frac{1(1-9)}{1-\sqrt{3}}$$

$= \frac{1((\sqrt{3})^4-1)}{\sqrt{3}-1}$

$= \frac{1(9-1)}{\sqrt{3}-1} = \frac{8}{\sqrt{3}-1}$

rationalising

$$\frac{8(\sqrt{3}+1)}{2}$$

$$= 4(\sqrt{3}+1)$$

(D)

34. Let $A = \emptyset$

$$P(A) = 2^n$$

$$= 2^0$$

$$= 1$$

(B)

35. $a = 1$

$$r = \frac{3}{3} = 3$$

hence $T_8 = ar^8$

$$= 1(3)^8$$

$$= 27 \times 9$$

$$= 243$$

(A)

36. $2x^2 - 7x + 5 = 0;$

$$\alpha^3 + \beta^3$$

First let us simplify

$$\alpha^3 + \beta^3$$

$$= (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$\alpha = -\frac{b}{a}$$

$$\beta = \frac{c}{a}$$

Hence;

$$\alpha = -\frac{(-7)}{2} = \frac{7}{2}$$

$$\beta = \frac{5}{2}$$

thus;

$$\left(\frac{7}{2}\right)^3 - \left[2\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\right]$$

$$= \frac{343}{8} - \frac{35}{2}$$

$$\frac{343 - 140}{8} = \frac{203}{8}$$

37. $\int_n = \frac{n}{2} (2a + (n-1)d)$

(A)

38. C	$= \frac{5}{2} \times 6$
39. All Term	
$T_4 = ar^3$	$= 15$
$a = 6, r = 3$	$=$
$= 6 \times (3)^3$	42. $S_n = 5n^2 - 3n$ — (1)
$= 6 \times 27$	at $n = 10$
$= 162$	at $T_n = a + (n-1)d$ — (2)
(D)	If n was assumed to be 1, at
40. $S_n = \frac{n}{2}(a+l)$	equ. (1)
$S_1 = \frac{6}{2}(2+20)$	$= 5(1)^2 - 3(1)$
$= 3(22)$	$= 5 - 3$
$= 66$	$= 2$
(R)	hence, $a = 2$
41. $a = 3, l = 3$	at $T_1, a = 2$
$S_n = \frac{n}{2}(a+l)$	hence, at $n = 2$, at equ. (1)
at $n = 5$	$S_2 = 5n^2 - 3n$
$S = \frac{5}{2}(3+3)$	$= 5(2)^2 - 3(2)$
	$= 5 \times 4 - 6$
	$= 20 - 6$
	$= 14$
	$T_2 = a + d$

$$\therefore S_{10} = a + atd$$

$$= 2a + d = 14$$

$$\text{hence, } 4 + d = 14$$

$$d = 14 - 4$$

$$d = 10$$

Hence, the 10th term

$$= a + 9d$$

$$= 2 + 90$$

$$= 92$$

(A)

$$43. \quad x/p^2 : a, b, c, xp^2$$

$$a = x/p^2$$

$$ar^4 = xp^2$$

$$x/p^2 \cdot r^4 = xp^2$$

$$r^4 = \frac{xp^2 \cdot x}{p^2}$$

$$r^4 = p^4$$

$$\text{hence, } r = p$$

$$a = ar$$

$$x/p^2 = xp$$

$$a = xp/p^2 = x/p$$

$$b = ar^2$$

$$b = x/p^2 \times p^2$$

$$b = x$$

$$c = ar^3$$

$$c = x/p^2 \times p^3$$

$$c = xp$$

hence, the geometric means are

$$x/p, x, xp$$

(B)

$$44. \quad x^2 - 2x + 1 = 0$$

$$\alpha = -b/a \quad \beta = c/a$$

$$\alpha = -(-2)/1 \quad ; \quad \beta = 1/1$$

$$\alpha = 2 \quad ; \quad \beta = 1$$

Hence, true and real roots

(C)

45. $2x^2 - 7x + 5 = 0$

$(x^2 + \beta^2)$ is + b found

; well first we simplify

the expression

$$(x^2 + \beta^2) = (x + \beta)^2 - 2\alpha\beta$$

$$x + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$x + \beta = \frac{-(-7)}{2} = \frac{7}{2}$$

$$\alpha\beta = \frac{5}{2}$$

$$x^2 + \beta^2 = \left(\frac{7}{2}\right)^2 - 2\left(\frac{5}{2}\right)$$

$$= \frac{49}{4} - \frac{5}{1}$$

$$\frac{49 - 20}{4} = \frac{29}{4}$$

(c)

46. $P(k+1) : P_n = n^3(n+1)$

$$(k+1)^3 (k+1+1)$$

$$= (k+1)^3 (k+2)$$

(k)

47. $x + iy : a, b, c, 5x + 3y$

$$a = x + iy$$

$$b = a + 4d$$

$$5x + 3y = x + iy + 4d$$

$$4d = 5x + 3y - x + iy$$

$$4d = 4x + 10y$$

$$d = \frac{4x + 10y}{4}$$

$$a = a + d$$

$$a = x + iy + \frac{4x + 10y}{4}$$

$$a = \frac{2(x + iy) + 4x + 10y}{4}$$

$$a = \frac{4x + 28y + 4x + 10y}{4}$$

$$a = \frac{8x + 38y}{4}$$

$$b = a + 2d$$

$$b = x + iy + 2\left(\frac{4x + 10y}{4}\right)$$

$$= \frac{2(x + iy) + (4x + 10y)}{2}$$

$\frac{2x+14y+4x+10y}{2}$	$a = a+d$
$b = \frac{6x+24y}{2}$	$a = x+y+x-y$
$b = 3x+12y$	$a = 2x+6y$
$c = a+3d$	$b = a+2d$
$c = x+y + \frac{3(4x+10y)}{4}$	$= x+y+2(x-y)$
$c = \frac{4(x+y) + 3(4x+10y)}{4}$	$x+y+2x-2y$
$c = \frac{4x+28y+12x+30y}{4}$	$3x+5y$
$c = \frac{16x+58y}{4}$	$c = a+3d$
$5x+3y = x+y+4d$	$= x+y+3(x-y)$
$4d = 5x-x+3y-y$	$= x+y+3x-3y$
$4d = 4x-4y$	$4x+4y$
$4d = \frac{4(x-y)}{4}$	The arithmetic mean is
$d = x-y$	$2x+6y; 3x+5y; 4x+4y.$
	(A)

$5x+3y = x+y+4d$	48. $3; a; b; c; d; e; 12$
$4d = 5x-x+3y-y$	$T_7 = a+6d$
$4d = 4x-4y$	$12 = 3+6d$
$4d = \frac{4(x-y)}{4}$	$6d = 9$
$d = x-y$	$d = \frac{9}{6}$
	$d = \frac{3}{2}$

$$a = a + d$$

$$= 3 + \frac{3}{2}$$

$$= \frac{6+3}{2} = \frac{9}{2} = 4.5$$

$$b = a + 2d$$

$$= 3 + 3$$

$$= 6$$

$$c = a + 3d$$

$$= 3 + \frac{9}{2}$$

$$= \frac{6+9}{2} = \frac{15}{2} = 7.5$$

$$d = a + 4d$$

$$= 3 + 6$$

$$= 9$$

$$e = a + 5d$$

$$= 3 + \frac{15}{2}$$

$$= \frac{6+15}{2} = \frac{21}{2} = 10.5$$

$$4.5, 6, 7.5, 9, 10.5$$

approx.

$$5, 6, 8, 9, 11$$

49. $P(n)$ is true for integers
(B.)

50. Same as no. 48.

$$4.5, 6, 7.5, 9, 10.5$$

Partly (B)

$$51. 2x^2 - 3x + 5 = 0$$

$$\underbrace{\hspace{2cm}}_{2x^2}$$

$$\underbrace{\hspace{2cm}}_{2x - 5x}$$

$$2x^2 - 5x$$

$$2x^2 + 2x - 5x + 5 = 0$$

$$2x(x+1) - 5(x-1) = 0$$

$$(2x-5)(x+1)(x-1)$$

here roots cannot be determined

(b).

52. $ax^2 + bx + c = 0$

$a \neq 0 ; b > 0$

$D = b^2 - 4ac$

$D > 0 ;$ Roots are real and Equal. (R)

53. $P = \{x : x \leq 3\}; Q = \{x : 5 < x < 12\}$

$R = \{x : -2 \leq x \leq 5\}$

$P = \{0, 1, 2, 3\}$

$Q = \{6, 7, 8, 9, 10, 11\}$

$R = \{-2, -1, 0, 1, 2, 3, 4, 5\}$

$Q \cup R = \{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

$P \cap (Q \cup R) = \{0, 1, 2, 3\}$

$P \cap (Q \cup R) = -4 \leq x \leq 3$

(R)

54. Workings on no 33;

Ans $4(\sqrt{3} + 1)$ (B)

55. $a = 2, c = 50$

$d = 4$

$L = a + (n-1)d$

$50 = 2 + (n-1)4$

$50 = 2 + 4n - 4$

$50 = 4n - 2$

$50 + 2 = 4n$

$4n = 52$

$n = \frac{52}{4}$

$n = 13$

(E)

56. $\frac{a+b}{2} = 18$

$\frac{12+b}{2} = 18$

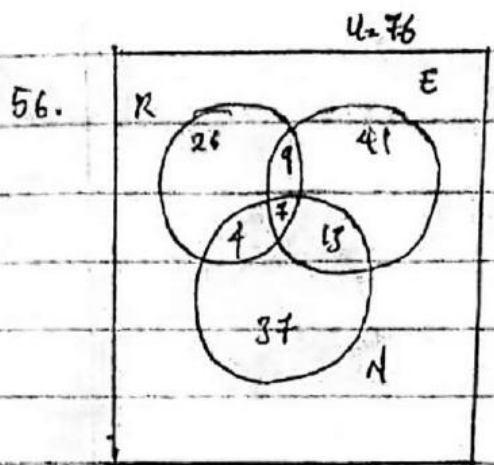
2

$b = 2(18) - 12$

$b = 36 - 12$

$b = 24$

(B)



$$n(N) = 37 - (4 + 7 + 13)$$

$$= 37 - (24)$$

$$= 13$$

but $a + a + d = S_n$ (at $n=2$)

$$2a + d = 14$$

$$2(5) + d = 14$$

$$d = 14 - 10$$

$$d = 4$$

$$T_n = a + (n-1)d$$

$$T_n = 5 + (n-1)4$$

$$= 5 + 4n - 4$$

$$= 4n + 1 \quad \textcircled{A}$$

57. First n term

$$S_n = 2n^2 + 3n$$

for $n=1$

$$S_1 = 2(1)^2 + 3(1)$$

$$= 5$$

and $a = 5$ having the first term

at $n=2$

$$S_2 = 2(2)^2 + 3(2)$$

$$= 2(4) + 6$$

$$= 14$$

58. $P(n+1) = P(n) = n^3(n+1)$

working in no 46.

Ans: $(n+1)^3(n+2) \quad \textcircled{A}$

59. $a = 32$

$$r = \frac{1}{2}$$

$$S_\infty = \frac{a}{1-r}$$

$$= \frac{32}{1-\frac{1}{2}} = \frac{32}{\frac{1}{2}} = 64 \quad \textcircled{A}$$

60. Workings at no 44.

Answer

$$2x+6y; 3x+5y; 4x+4y$$

(A)

65. Please refer workings to no 24.

Answer = 6 students

(B)

61. Answer equals: $-\frac{3}{2}$ (D)

Workings at no 46.

66. Please refer workings to no 32.

Answer = 8

(C)

62. Workings at no 35.

Answer = 243 (C)

$$67. \frac{n(n+1)}{n-2}; a; \frac{n(n+1)^T}{(n-2)^3}$$

$$T_1; T_2; T_3$$

$$T_3 = ar^2$$

$$T_1 = a$$

$$\frac{n(n+1)^T}{(n-2)^3} = \frac{n(n+1)}{(n-2)} r^2$$

$$r^2 = \frac{n(n+1)^2}{(n-2)^2}$$

$$r = \sqrt{\frac{n(n+1)^2}{(n-2)^2}}$$

$$r = \frac{n+1}{n-2}$$

63. Workings at no 24.

$$\text{Answer} = \frac{2}{625}$$

(E)

64. $a=10; b=20; c=30$

$$a.m = \frac{a+b}{2}$$

$$= \frac{10+20}{2} = \frac{30}{2} = 15$$

(A)

$$\begin{aligned}
 I_2 &= ar \\
 &= \frac{a(x+1)}{x-2} \left(\frac{(x+1)}{x-2} \sqrt{x} \right) \\
 &= \frac{x(x+1)^2 \sqrt{x}}{(x-2)^2} \\
 &= \text{Error}
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= ar \\
 &= \frac{x(x+1)}{(x-2)} \cdot \frac{(x+1)}{(x-2)} \\
 &= \frac{x(x+1)^2}{(x-2)^2}
 \end{aligned}$$

68. $ar = 9$ --- (i)

$ar^3 = 81$ --- (ii)

Divide eqn. (ii) by (i)

$$r^2 = 9$$

$$r = \sqrt{9} = 3$$

$$ar = 9$$

$$3a = 9$$

$$a = \frac{9}{3}$$

$$a = 3$$

(A)

69. $a = 2, r = 20$

Refer working to no 40

Answer is 66 (B)

70. C

71. $243; a; b; c; d; -1$

$$-1 = r$$

$$a + 5d = -1$$

$$a = 243$$

$$243 + 5d = -1$$

$$5d = -244$$

$$d = \frac{-244}{5}$$

$$a = a + d$$

$$= 243 + \frac{244}{5}$$

$$= 243 - 48.8$$

$$= 194.2$$

$$b = a + 2d$$

$$= 243 - 97.6$$

$$= 145.4$$

$c = a + 3d$ $c = 243 - 146 - 9$ $c = 96 - 6$	75. Refer workings to no 53. Answer equals: $-4 \leq n \leq 3$ (B)
$d = a + 4d$ $= 243 - 195 - 2$ $= 47 - 8$ Arithmetic means are $194, 195, 197, 198.$	76. Refer workings to no 57. Answer equals $4n + 1$ (B)
72. Refer workings to no 23. Answer is 2048 (B)	77. Refer workings to no 46. Answer equals $(k+1)^3 (k+2)$ Further simplifying, $(k+1)^2 (k+1)(k+2)$
73. Refer workings to no 67. Answer = $\frac{n(n+1)^2}{(n-2)^2}$ (B)	78. Refer workings to no 59. Answer equals 64 (A)
74. $P(n) = n^2(n+3)(n^2-1)$ $P(k+1) = (k+1)^2(k+1+3)((k+1)^2-1)$ $(k+1)^2(k+4)(k(k+2))$ hence, $k(k+1)^2(k+4)(k+2)$	79. Refer workings to no 47a Answer equals $2x+6y, 3x+5y, 4x+4y$

80. Refer workings to no 16

Answer equals $-3/2$

81. Refer workings to no 45

Answer question = 243

82. Refer workings to no 29.

Answer = $2/625$

83. Refer workings to no 64

Answer equals

15. (B)

84. Refer workings to no 27.

Answer = 6 students

(B)

85. Refer workings to no 32

Answer = 8

(B)

86. $ar = 35$ — (i)

$ar^3 = 875$ — (ii)

Divide equ. (ii) by (i)

$$r^2 = 25$$

$$r = \sqrt{25}$$

$$r = 5$$

Find the Third Term

$$T_3 = ar^2$$

First to find a

$$ar = 35 \text{ from (i)}$$

$$5a = 35$$

$$a = 7$$

$$\text{Hence, } ar^2 = 7 \times 25 \\ = 175$$

87. Refer workings to no 43.

Answer equals; x/p ; x ; x/p

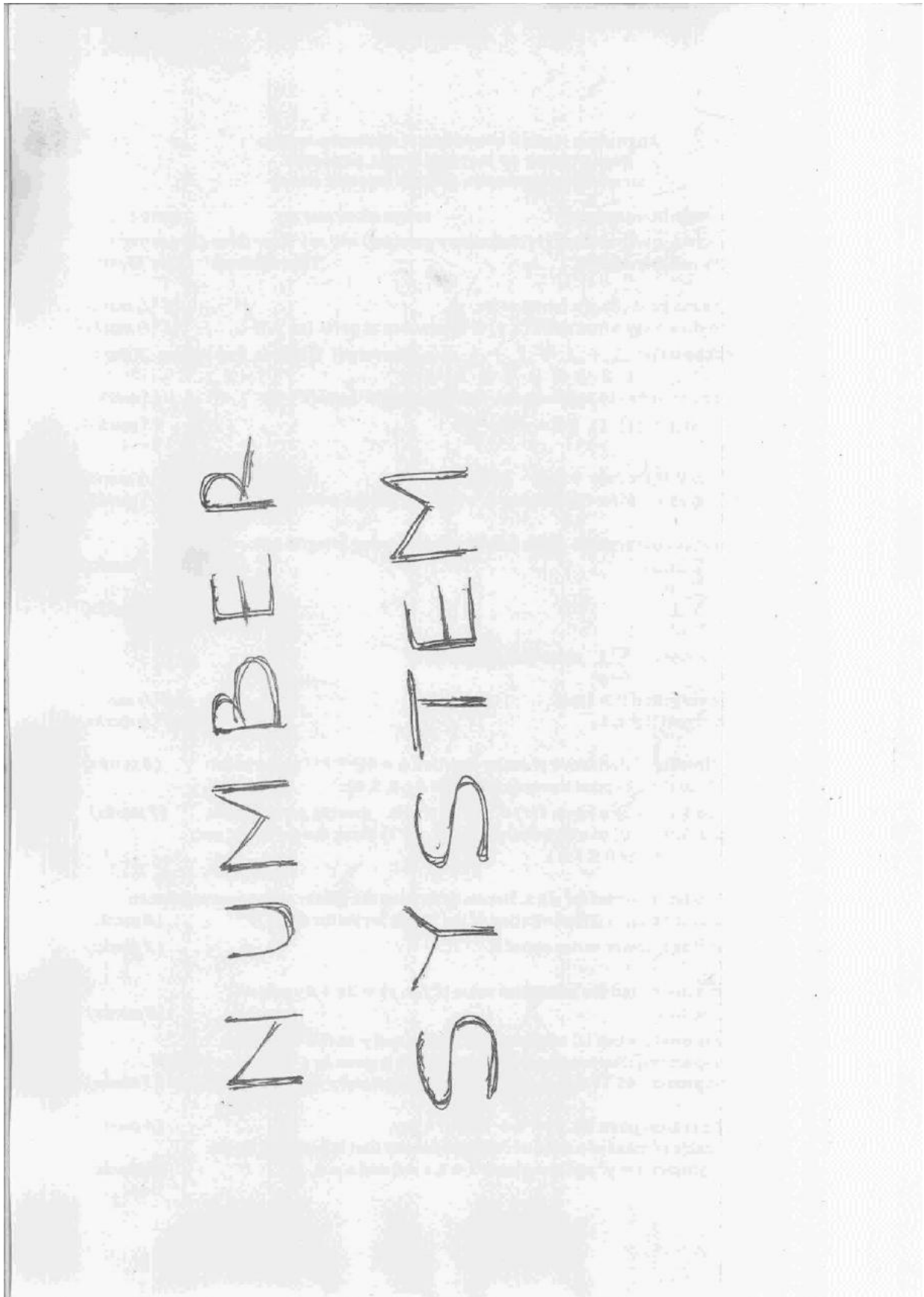
88. $T_7 = ar^6 = 4374$ — (i)

$T_5 = ar^4 = 162$ — (ii)

Divide equ. (i) by (ii)

Paper 4 of 5 — Cyssa hub100LVL materials1st SemesterMTH 101NUMBER SYSTEM 1 - 10.pdf

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1.0 HISTORICAL INTRODUCTION ^①

Numbers form part of the divine gifts of God to the human race. They help man to solve the problems of identifying, ordering, classifying and differentiating objects by quantity, quality, shape, size, weight, colour and composition.

Numbers are thus the basic building blocks of development and civilization. Without numbers, no form of development can be achieved.

The theory of numbers is that branch of Mathematics which deals with properties of the whole numbers,

1, 2, 3, 4, 5, 6, ...

also called the counting numbers, or positive integers.

The positive integers are undoubtedly man's first mathematical creation. It is hardly possible to imagine human beings without the ability to count, at least within a limited range. Historical record shows that as early as 5700 BC the ancient Sumerians kept a calendar, so they must have developed some form of arithmetic.

By 2500 BC the Sumerians had developed a number system using 60 as a base. This was passed on to the Babylonians,

(2)

who became highly skilled calculators.

Babylonian clay tablets containing elaborate mathematical tables have been found, dating back to 2000 BC.

When ancient civilizations reached a level which provided leisure time to ponder about things, some people began to speculate about the nature and the properties of numbers. This curiosity developed into a sort of number mysticism or numerology.

Although different civilizations developed number systems with unique counting units (number bases) and symbols (numerals) for recording numbers and number units, the basic concept of number is universally the same.

The Hindu-Arabic System which is popularly used in writing and computation is a base ten system which comprises ten numbers (digits): 1, 2, 3, 4, 5, 6, 7, 8, 9, 0. Worth noting is the Hindu-Arabic number base system. Base ten (10) system is widely used today.

The use of numbers in sorting, classifying, comparing, combining, ordering, measuring and computing quantities gave rise to the development of number systems based on fundamental axioms, properties, principles, and laws with the accompanying symbols and notations.

All the number systems are off-springs of the counting numbers denoted by $\mathbb{N}$.

③

Natural numbers, $\mathbb{N}$ comprise $1, 2, 3, 4, 5, 6, \dots$

The counting numbers are also called the Natural numbers. Thus, $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$.

The introduction of the number 0 (zero) expanded the counting number system to obtain the Whole numbers denoted by $\mathbb{W}$ and thus given as $\mathbb{W} = \{0, 1, 2, 3, 4, 5, \dots\}$.

When negative whole numbers are added to $\mathbb{W}$ (i.e. positive whole number), we obtain a set of numbers called Integer denoted by $\mathbb{I}$ or $\mathbb{Z}$ and this is made up of:

(i) the set of positive integers: $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$

(ii) the number Zero (0) and

(iii) the set of negative integers: $\mathbb{Z}^- = \{\dots, -3, -2, -1\}$

Thus, $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \pm 4, \dots\}$.

NB: The set of whole numbers is also referred to as the set of non-negative integers.

Meanwhile, the set of integers can only accommodate operations of addition, subtraction and multiplication. Division could only be performed for some pairs of numbers. Thus, further extension of the set of integers to include quotients produced a new set of numbers called rational (also called fractional) numbers denoted by $\mathbb{Q}$ and given by

$$\mathbb{Q} = \left\{ \frac{m}{n} : m \text{ and } n \text{ are integers and } n \neq 0 \right\}.$$

(4)

Moreover, in order to solve most algebraic equations involving roots of positive numbers like $\sqrt{2}$, $\sqrt{3}$, $\sqrt{8}$, $\sqrt{18}$ etc, another extension of numbers called irrational numbers was made.

Thereafter, numbers like e , π , e^π etc were introduced. They are called Transcendental numbers. Later in the history of number theory, in order to perform scientific and technological computation new set of numbers was added to enrich our number system. This new set of numbers consisting the natural numbers, the integers, the rational numbers, irrational numbers and the transcendental numbers is called the real numbers and denoted by $\mathbb{R}$.

The inclusion of numbers arising from roots of negative numbers produced new set of numbers called complex numbers and other higher level number systems.

The set of complex numbers is denoted by $\mathbb{C}$ and compactly expressed as $\mathbb{C} = a + ib$ where, a is the real part, b is the imaginary part and i is the imaginary number (i.e. $i = \sqrt{-1}$).

Thus, $\mathbb{C}$ contains $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{W}$ and $\mathbb{N}$

$\mathbb{R}$ contains $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{W}$ and $\mathbb{N}$

$\mathbb{Q}$ contains $\mathbb{Z}$, $\mathbb{W}$ and $\mathbb{N}$

$\mathbb{Z}$ contains $\mathbb{W}$ and $\mathbb{N}$

$\mathbb{W}$ contains $\mathbb{N}$

and $\mathbb{N} = \{1, 2, 3, 5, 6, \dots\}$

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Do not
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this margin(a) Natural Number ($\mathbb{N}$)

- This is a set of counting numbers

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

- The even natural numbers is the set

$$\{2K : K \in \mathbb{N}\} = \{2, 4, 6, 8, \dots\}$$

- The odd natural numbers is the set

$$\{2K-1 : K \in \mathbb{N}\} = \{1, 3, 5, 7, 9, \dots\}$$

- A Prime number is a natural number which has two (2) factors / divisors namely 1 and itself

- The set of prime numbers is the set

$$\mathbb{P} = \{2, 3, 5, 7, 11, 13, 17, 19, 23, \dots\}$$

- A composite number is a natural number which can be expressed as a product of two (2) natural numbers, each of which is not equal to 1 or itself.

The set of composite numbers is the set

$$\{4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, \dots\}$$

(b) Integers ($\mathbb{I}$ or $\mathbb{Z}$)

The set of integers is the set

$$\mathbb{Z} \text{ or } \mathbb{I} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

Integers comprise even, odd, prime and composite numbers.

(c) Rational numbers ($\mathbb{Q}$)

The set of rational number is the set

$$\mathbb{Q} = \{P/Q : P, Q \in \mathbb{Z}, Q \neq 0\}$$

Rational numbers consist of all the integers and all fractions

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(d) Real Numbers ($\mathbb{R}$)

The system of rational numbers, extended to a larger system is known as the system of real numbers, which constitute rational as well as irrational numbers.

The set of real numbers is the set;
 $\mathbb{R} = \mathbb{Q} \cup \{\text{Irrational numbers}\}$

(e) Complex Numbers ($\mathbb{C}$)

By the definition, a complex number is any number z that can be expressed in the form:

$$z = x + iy$$

where, x and y are real numbers and i is an imaginary number, given as; $i = -1$

(f) Then, we have the following inclusions;

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

In Complex number $\mathbb{C}$, the solution of the equation, $x^2 + 1 = 0$

$$\therefore x^2 = -1$$

$$x = \pm\sqrt{-1}$$

$$\Rightarrow x = \pm i$$

ALGEBRA OF COMPLEX NUMBERS

Let z_1 and z_2 be any two complex numbers given as;

$$z_1 = a + ib \quad \text{and} \quad z_2 = c + id$$

(i) Equality of Complex numbers

Complex numbers, z_1 and z_2 are said to be equal, if their real parts and imaginary parts are equal.

$$\text{i.e. } a = c \quad \text{and} \quad b = d$$

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	(ii) Addition and Subtraction of Complex numbers	
	The sum of Z_1 and Z_2 are defined as a complex number Z , s.t	
	$Z = Z_1 + Z_2$	
	$\Rightarrow Z = a+ib + c+id = (a+c) + i(b+d)$	
	Similarly, for subtraction,	
	$Z = Z_1 - Z_2$	
	$\Rightarrow Z = (a+ib) - (c+id) = (a-c) + i(b-d)$	
	Example 1	
	Let $Z_1 = 5+3i$, $Z_2 = 2+7i$ and $Z_3 = 6-5i$	
	Find:	
	(a) $Z_1 + Z_2$ (b) $Z_3 + Z_2$ (c) $Z_1 - Z_3$ (d) $Z_2 - Z_1$	
	Solution	
	(a) $Z_1 + Z_2 = (5+3i) + (2+7i) = 7+10i$	
	(b) $Z_3 + Z_2 = (6-5i) + (2+7i) = 8+2i$	
	(c) $Z_1 - Z_3 = (5+3i) - (6-5i) = -1+8i$	
	(d) $Z_2 - Z_1 = (2+7i) - (5-3i) = -3+10i$	
	(iii) Multiplication of Complex numbers, $Z_1 \& Z_2$	
	$Z_1 \cdot Z_2 = (a+ib)(c+id)$	
	$Z_1 Z_2 = ac + iad + ibc + i^2 bd$	
	$\Rightarrow Z_1 Z_2 = ac + i(ad+bc) - bd$	
	$= (ac-bd) + i(ad+bc)$	
	(iv) Complex Conjugate	
	Consider a complex number $Z = a+ib$	
	The conjugate of Z is denoted and defined by	
	$\bar{Z} = \overline{a+ib} = a-ib$	
	e.g conjugate of $Z = 3+2i$ is $\bar{Z} = 3-2i$	
	" " $Z = 7-5i$ is $\bar{Z} = 7+5i$	

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(V) Absolute value of complex number.

The absolute value of $Z = a + ib$ denoted by $|Z|$, is defined to be non-negative square root of $Z\bar{Z}$ namely, $|Z| = \sqrt{Z\bar{Z}} = \sqrt{a^2 + b^2}$

(VI) Inverse of Z .

Suppose $Z \neq 0$, the inverse Z^{-1} of Z is given by:

$$Z^{-1} = \frac{1}{Z} = \frac{1}{a+ib} = \frac{1}{a+ib} \times \frac{a-ib}{a-ib}$$

$$\Rightarrow Z^{-1} = \frac{a-ib}{a^2 - i^2 b^2} = \frac{a-ib}{a^2 + b^2} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i$$

(VII) Division of complex numbers Z_1 and Z_2

Let $Z_1 = a + ib$ and $Z_2 = c + id$

$$\frac{Z_1}{Z_2} = \frac{a+ib}{c+id} = \frac{a+ib}{c+id} \times \frac{c-id}{c-id}$$

$$= \frac{(ac+bd) + (bc-ad)i}{c^2 + d^2}$$

$$\Rightarrow \frac{Z_1}{Z_2} = \frac{ac+bd}{c^2+d^2} + \frac{(bc-ad)i}{c^2+d^2}$$

Example 2

Suppose $Z_1 = 5 + 3i$, $Z_2 = 2 + 7i$ and $Z_3 = 6 - 5i$,

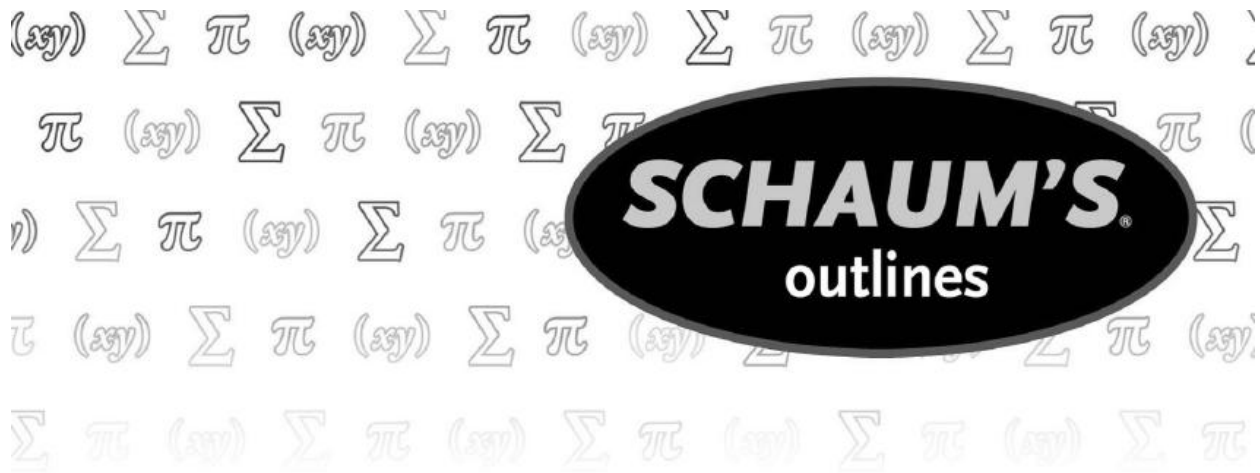
(a) $Z_1 Z_2$ (b) $Z_2 Z_3$ (c) $Z_1^2 - 5Z_3$ (d) Conjugate of Z_3 (e) " " Z_1 (f) Absolute value of Z_2 (g) absolute " " Z_3 (h) Z_2^{-1} (i) Z_3^{-2} (j) Z_2/Z_1 (k) Z_1/Z_3 (l) $\frac{Z_1^2}{Z_1 Z_2}$

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	SOLUTION	
	Given, $Z_1 = 5+3i$, $Z_2 = 2+7i$ and $Z_3 = 6-5i$	
	(a) $Z_1 Z_2 = (5+3i)(2+7i)$ $= 10 + 35i + 6i + 21i^2$ $= 10 - 21 + 41i = -11 + 41i$	
	(b) $Z_2 Z_3 = (2+7i)(6-5i)$ $= 12 - 10i + 42i + 35 = 47 + 32i$	
	(c) $Z_1 - 5Z_3 = (5+3i)(5+3i) - 5(6-5i)$ $= 25 + 30i - 9 - 30 + 25i$ $= -14 + 55i$	
	(d) Conjugate of Z_3 $\Rightarrow \overline{Z_3} = \overline{6-5i} = 6+5i$	
	(e) Conjugate of Z_1 $\Rightarrow \overline{Z_1} = \overline{5+3i} = 5-3i$	
	(f) Absolute value of Z_2 $\Rightarrow Z_2 = \sqrt{2^2+7^2} = \sqrt{4+49} = \sqrt{53}$	
	(g) Absolute value of Z_3 $\Rightarrow Z_3 = \sqrt{6^2+(-5)^2} = \sqrt{36+25} = \sqrt{61}$	
	(h) $Z_2^{-1} = \frac{1}{2+7i} = \frac{1}{2+7i} \times \frac{2-7i}{2-7i}$ $= \frac{2-7i}{2^2+7^2} = \frac{2-7i}{4+49} = \frac{2-7i}{53}$ $\Rightarrow Z_2^{-1} = \frac{2}{53} - \frac{7i}{53}$	

write in this margin	QUESTION NO. (10)	write in this margin
	$(i) \quad Z_3^{-2} = \frac{1}{Z_3^2} = \frac{1}{(6-5i)(6-5i)}$ $\Rightarrow Z_3^{-2} = \frac{1}{6^2+5^2} = \frac{1}{61}$	
	$(j) \quad \frac{Z_2}{Z_1} = \frac{2+7i}{5+3i}$ $= \frac{2+7i}{5+3i} \times \frac{5-3i}{5-3i}$ $= \frac{10-6i+35i+21}{5^2+3^2} = \frac{31+29i}{34}$ $\Rightarrow Z_2/Z_1 = 31/34 + 29/34i$	
	$(k) \quad \frac{Z_1}{Z_3} = \frac{5+3i}{6-5i}$ $= \frac{5+3i}{6-5i} \times \frac{6+5i}{6+5i} = \frac{30+43i-15}{36+25}$ $\Rightarrow \frac{Z_1}{Z_3} = \frac{15+43i}{61} = \frac{15}{61} + \frac{43}{61}i$	
	$(l) \quad \frac{Z_3^2}{Z_1 Z_2} = \frac{(6-5i)(6-5i)}{(5+3i)(2+7i)} = \frac{36-25}{10+35i+6i-21}$ $= \frac{11}{-11+41i} = \frac{11}{-11+41i} \times \frac{-11-41i}{-11-41i}$ $\Rightarrow \frac{Z_3^2}{Z_1 Z_2} =$	
	Exercise Express in the form $a + ib$.	
	$(a) (2+4i)+(5-2i)$ $(c) (4+3i)(2+6i)$ $(e) \frac{2+3i}{3+2i}$ $(g) \frac{2+3i}{i(4-5i)} + \frac{2}{i}$ $(b) (1-8i)-(7+2i)$ $(d) (1+3i)^{-1}$ $(f) \left[\frac{5(1+i)}{1+3i} \right]^2$ $(h) i^3(1+i) + i^5(3-i) + i^7(2+i)$	

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Preface

The purpose of this book is to help students understand and use the calculus. Everything has been aimed toward making this easier, especially for students with limited background in mathematics or for readers who have forgotten their earlier training in mathematics. The topics covered include all the material of standard courses in elementary and intermediate calculus. The direct and concise exposition typical of the Schaum Outline series has been amplified by a large number of examples, followed by many carefully solved problems. In choosing these problems, we have attempted to anticipate the difficulties that normally beset the beginner. In addition, each chapter concludes with a collection of supplementary exercises with answers.

This sixth edition has enlarged the number of solved problems and supplementary exercises. Moreover, we have made a great effort to go over ticklish points of algebra or geometry that are likely to confuse the student. The author believes that most of the mistakes that students make in a calculus course are not due to a deficient comprehension of the principles of calculus, but rather to their weakness in high-school algebra or geometry. Students are urged to continue the study of each chapter until they are confident about their mastery of the material. A good test of that accomplishment would be their ability to answer the supplementary problems.

The author would like to thank many people who have written to me with corrections and suggestions, in particular Danielle Cinq-Mars, Lawrence Collins, L.D. De Jonge, Konrad Duch, Stephanie Happ, Lindsey Oh, and Stephen B. Soffer. He is also grateful to his editor, Charles Wall, for all his patient help and guidance.

ELLIOTT MENDELSON

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CHAPTER 1

Linear Coordinate Systems. Absolute Value. Inequalities

Linear Coordinate System

A linear coordinate system is a graphical representation of the real numbers as the points of a straight line. To each number corresponds one and only one point, and to each point corresponds one and only one number.

To set up a linear coordinate system on a given line: (1) select any point of the line as the *origin* and let that point correspond to the number 0; (2) choose a positive direction on the line and indicate that direction by an arrow; (3) choose a fixed distance as a unit of measure. If x is a positive number, find the point corresponding to x by moving a distance of x units from the origin in the positive direction. If x is negative, find the point corresponding to x by moving a distance of $-x$ units from the origin in the negative direction. (For example, if $x = -2$, then $-x = 2$ and the corresponding point lies 2 units from the origin in the negative direction.) See Fig. 1-1.

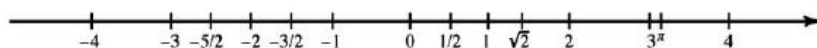


Fig. 1-1

The number assigned to a point by a coordinate system is called the *coordinate* of that point. We often will talk as if there is no distinction between a point and its coordinate. Thus, we might refer to “the point 3” rather than to “the point with coordinate 3.”

The absolute value $|x|$ of a number x is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \text{ is zero or a positive number} \\ -x & \text{if } x \text{ is a negative number} \end{cases}$$

For example, $|4| = 4$, $|-3| = -(-3) = 3$, and $|0| = 0$. Notice that, if x is a negative number, then $-x$ is positive. Thus, $|x| \geq 0$ for all x .

The following properties hold for any numbers x and y .

- (1.1) $|-x| = |x|$
When $x = 0$, $|-x| = |-0| = |0| = |x|$.
When $x > 0$, $-x < 0$ and $|-x| = -(-x) = x = |x|$.
When $x < 0$, $-x > 0$, and $|-x| = -x = |x|$.
- (1.2) $|x - y| = |y - x|$
This follows from (1.1), since $y - x = -(x - y)$.
- (1.3) $|x| = c$ implies that $x = \pm c$.
For example, if $|x| = 2$, then $x = \pm 2$. For the proof, assume $|x| = c$.
If $x \geq 0$, $x = |x| = c$. If $x < 0$, $-x = |x| = c$; then $x = -(-x) = -c$.
- (1.4) $|x|^2 = x^2$
If $x \geq 0$, $|x| = x$ and $|x|^2 = x^2$. If $x \leq 0$, $|x| = -x$ and $|x|^2 = (-x)^2 = x^2$.
- (1.5) $|xy| = |x| \cdot |y|$
By (1.4), $|xy|^2 = (xy)^2 = x^2 y^2 = |x|^2 |y|^2 = (|x| \cdot |y|)^2$. Since absolute values are nonnegative, taking square roots yields $|xy| = |x| \cdot |y|$.

$$(1.6) \quad \left| \frac{x}{y} \right| = \frac{|x|}{|y|} \text{ if } y \neq 0$$

By (1.5), $|y| \left| \frac{x}{y} \right| = \left| y \cdot \frac{x}{y} \right| = |x|$. Divide by $|y|$.

$$(1.7) \quad |x| = |y| \text{ implies that } x = \pm y$$

Assume $|x| = |y|$. If $y = 0$, $|x| = |0| = 0$ and (1.3) yields $x = 0$. If $y \neq 0$, then by (1.6),

$$\left| \frac{x}{y} \right| = \frac{|x|}{|y|} = 1$$

So, by (1.3), $x/y = \pm 1$. Hence, $x = \pm y$.

$$(1.8) \quad \text{Let } c \geq 0. \text{ Then } |x| \leq c \text{ if and only if } -c \leq x \leq c. \text{ See Fig. 1-2.}$$

Assume $x \geq 0$. Then $|x| = x$. Also, since $c \geq 0$, $-c \leq 0 \leq x$. So, $|x| \leq c$ if and only if $-c \leq x \leq c$. Now assume $x < 0$. Then $|x| = -x$. Also, $x < 0 \leq c$. Moreover, $-x \leq c$ if and only if $-c \leq x$. (Multiplying or dividing an equality by a negative number reverses the inequality.) Hence, $|x| \leq c$ if and only if $-c \leq x \leq c$.

$$(1.9) \quad \text{Let } c \geq 0. \text{ Then } |x| < c \text{ if and only if } -c < x < c. \text{ See Fig. 1-2. The reasoning here is similar to that for (1.8).}$$



Fig. 1-2

$$(1.10) \quad -|x| \leq x \leq |x|$$

If $x \geq 0$, $x = |x|$. If $x < 0$, $|x| = -x$ and, therefore, $x = -|x|$.

$$(1.11) \quad |x + y| \leq |x| + |y| \text{ (triangle inequality)}$$

By (1.8), $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding, we obtain $-(|x| + |y|) \leq x + y \leq |x| + |y|$. Then $|x + y| \leq |x| + |y|$ by (1.8). [In (1.8), replace c by $|x| + |y|$ and x by $x + y$.]

Let a coordinate system be given on a line. Let P_1 and P_2 be points on the line having coordinates x_1 and x_2 . See Fig. 1-3. Then:

$$(1.12) \quad |x_1 - x_2| = P_1P_2 = \text{distance between } P_1 \text{ and } P_2.$$

This is clear when $0 < x_1 < x_2$ and when $x_1 < x_2 < 0$. When $x_1 < 0 < x_2$, and if we denote the origin by O , then $P_1P_2 = P_1O + OP_2 = (-x_1) + x_2 = x_2 - x_1 = |x_2 - x_1| = |x_1 - x_2|$.

As a special case of (1.12), when P_2 is the origin (and $x_2 = 0$):

$$(1.13) \quad |x_1| = \text{distance between } P_1 \text{ and the origin.}$$



Fig. 1-3

Finite Intervals

Let $a < b$.

The *open interval* (a, b) is defined to be the set of all numbers between a and b , that is, the set of all x such that $a < x < b$. We shall use the term *open interval* and the notation (a, b) also for all the points between the points with coordinates a and b on a line. Notice that the open interval (a, b) does not contain the *endpoints* a and b . See Fig. 1-4.

The *closed interval* $[a, b]$ is defined to be the set of all numbers between a and b or equal to a or b , that is, the set of all x such that $a \leq x \leq b$. As in the case of open intervals, we extend the terminology and notation to points. Notice that the closed interval $[a, b]$ contains both endpoints a and b . See Fig. 1-4.

CHAPTER 1 Linear Coordinate Systems

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Fig. 1-4

By a *half-open interval* we mean an open interval (a, b) together with one of its endpoints. There are two such intervals: $[a, b)$ is the set of all x such that $a \leq x < b$, and $(a, b]$ is the set of all x such that $a < x \leq b$.

Infinite Intervals

Let (a, ∞) denote the set of all x such that $a < x$.

Let $[a, \infty)$ denote the set of all x such that $a \leq x$.

Let $(-\infty, b)$ denote the set of all x such that $x < b$.

Let $(-\infty, b]$ denote the set of all x such that $x \leq b$.

Inequalities

Any inequality, such as $2x - 3 > 0$ or $5 < 3x + 10 \leq 16$, determines an interval. To solve an inequality means to determine the corresponding interval of numbers that satisfy the inequality.

EXAMPLE 1.1: Solve $2x - 3 > 0$.

$$2x - 3 > 0$$

$$2x > 3 \quad (\text{Adding } 3)$$

$$x > \frac{3}{2} \quad (\text{Dividing by } 2)$$

Thus, the corresponding interval is $(\frac{3}{2}, \infty)$.

EXAMPLE 1.2: Solve $5 < 3x + 10 \leq 16$.

$$5 < 3x + 10 \leq 16$$

$$-5 < 3x \leq 6 \quad (\text{Subtracting } 10)$$

$$-\frac{5}{3} < x \leq 2 \quad (\text{Dividing by } 3)$$

Thus, the corresponding interval is $(-\frac{5}{3}, 2]$.

EXAMPLE 1.3: Solve $-2x + 3 < 7$.

$$-2x + 3 < 7$$

$$-2x < 4 \quad (\text{Subtracting } 3)$$

$$x > -2 \quad (\text{Dividing by } -2)$$

(Recall that dividing by a negative number reverses an inequality.) Thus, the corresponding interval is $(-2, \infty)$.

SOLVED PROBLEMS

1. Describe and diagram the following intervals, and write their interval notation, (a) $-3 < x < 5$; (b) $2 \leq x \leq 6$; (c) $-4 < x \leq 0$; (d) $x > 5$; (e) $x \leq 2$; (f) $3x - 4 \leq 8$; (g) $1 < 5 - 3x < 11$.

(a) All numbers greater than -3 and less than 5 ; the interval notation is $(-3, 5)$:



- (b) All numbers equal to or greater than 2 and less than or equal to 6; $[2, 6]$:



- (c) All numbers greater than -4 and less than or equal to 0; $(-4, 0]$:



- (d) All numbers greater than 5; $(5, \infty)$:



- (e) All numbers less than or equal to 2; $(-\infty, 2]$:



- (f) $3x - 4 \leq 8$ is equivalent to $3x \leq 12$ and, therefore, to $x \leq 4$. Thus, we get $(-\infty, 4]$:



- (g) $1 < 5 - 3x < 11$
 $-4 < -3x < 6$ (Subtracting 5)
 $-2 < x < \frac{4}{3}$ (Dividing by -3 ; note the reversal of inequalities)

Thus, we obtain $(-2, \frac{4}{3})$:



2. Describe and diagram the intervals determined by the following inequalities, (a) $|x| < 2$; (b) $|x| > 3$; (c) $|x - 3| < 1$; (d) $|x - 2| < \delta$ where $\delta > 0$; (e) $|x + 2| \leq 3$; (f) $0 < |x - 4| < \delta$ where $\delta > 0$.

- (a) By property (1.9), this is equivalent to $-2 < x < 2$, defining the open interval $(-2, 2)$.



- (b) By property (1.8), $|x| \leq 3$ is equivalent to $-3 \leq x \leq 3$. Taking negations, $|x| > 3$ is equivalent to $x < -3$ or $x > 3$, which defines the union of the intervals $(-\infty, -3)$ and $(3, \infty)$.



- (c) By property (1.12), this says that the distance between x and 3 is less than 1, which is equivalent to $2 < x < 4$. This defines the open interval $(2, 4)$.



We can also note that $|x - 3| < 1$ is equivalent to $-1 < x - 3 < 1$. Adding 3, we obtain $2 < x < 4$.

- (d) This is equivalent to saying that the distance between x and 2 is less than δ , or that $2 - \delta < x < 2 + \delta$, which defines the open interval $(2 - \delta, 2 + \delta)$. This interval is called the δ -neighborhood of 2:



CHAPTER 1 Linear Coordinate Systems

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- (e) $|x + 2| < 3$ is equivalent to $-3 < x + 2 < 3$. Subtracting 2, we obtain $-5 < x < 1$, which defines the open interval $(-5, 1)$:



- (f) The inequality $|x - 4| < \delta$ determines the interval $4 - \delta < x < 4 + \delta$. The additional condition $0 < |x - 4|$ tells us that $x \neq 4$. Thus, we get the union of the two intervals $(4 - \delta, 4)$ and $(4, 4 + \delta)$. The result is called the *deleted δ -neighborhood of 4*:



3. Describe and diagram the intervals determined by the following inequalities, (a) $|5 - x| \leq 3$; (b) $|2x - 3| < 5$; (c) $|1 - 4x| < \frac{1}{2}$.

- (a) Since $|5 - x| = |x - 5|$, we have $|x - 5| \leq 3$, which is equivalent to $-3 \leq x - 5 \leq 3$. Adding 5, we get $2 \leq x \leq 8$, which defines the closed interval $[2, 8]$:



- (b) $|2x - 3| < 5$ is equivalent to $-5 < 2x - 3 < 5$. Adding 3, we have $-2 < 2x < 8$; then dividing by 2 yields $-1 < x < 4$, which defines the open interval $(-1, 4)$:



- (c) Since $|1 - 4x| = |4x - 1|$, we have $|4x - 1| < \frac{1}{2}$, which is equivalent to $-\frac{1}{2} < 4x - 1 < \frac{1}{2}$. Adding 1, we get $\frac{1}{2} < 4x < \frac{3}{2}$. Dividing by 4, we obtain $\frac{1}{8} < x < \frac{3}{8}$, which defines the open interval $(\frac{1}{8}, \frac{3}{8})$:



4. Solve the inequalities: (a) $18x - 3x^2 > 0$; (b) $(x + 3)(x - 2)(x - 4) < 0$; (c) $(x + 1)^2(x - 3) > 0$, and diagram the solutions.

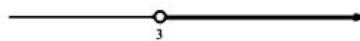
- (a) Set $18x - 3x^2 = 3x(6 - x) = 0$, obtaining $x = 0$ and $x = 6$. We need to determine the sign of $18x - 3x^2$ on each of the intervals $x < 0$, $0 < x < 6$, and $x > 6$, to determine where $18x - 3x^2 > 0$. Note that it is negative when $x < 0$ (since x is negative and $6 - x$ is positive). It becomes positive when we pass from left to right through 0 (since x changes sign but $6 - x$ remains positive), and it becomes negative when we pass through 6 (since x remains positive but $6 - x$ changes to negative). Hence, it is positive when and only when $0 < x < 6$.



- (b) The crucial points are $x = -3$, $x = 2$, and $x = 4$. Note that $(x + 3)(x - 2)(x - 4)$ is negative for $x < -3$ (since each of the factors is negative) and that it changes sign when we pass through each of the crucial points. Hence, it is negative for $x < -3$ and for $2 < x < 4$:



- (c) Note that $(x + 1)$ is always positive (except at $x = -1$, where it is 0). Hence $(x + 1)^2(x - 3) > 0$ when and only when $x - 3 > 0$, that is, for $x > 3$:



5. Solve $|3x - 7| = 8$.

By (1.3), $|3x - 7| = 8$ if and only if $3x - 7 = \pm 8$. Thus, we need to solve $3x - 7 = 8$ and $3x - 7 = -8$. Hence, we get $x = 5$ or $x = -\frac{1}{3}$.

6. Solve $\frac{2x+1}{x+3} > 3$.

Case 1: $x+3 > 0$. Multiply by $x+3$ to obtain $2x+1 > 3x+9$, which reduces to $-8 > x$. However, since $x+3 > 0$, it must be that $x > -3$. Thus, this case yields no solutions.

Case 2: $x+3 < 0$. Multiply by $x+3$ to obtain $2x+1 < 3x+9$. (Note that the inequality is reversed, since we multiplied by a negative number.) This yields $-8 < x$. Since $x+3 < 0$, we have $x < -3$. Thus, the only solutions are $-8 < x < -3$.

7. Solve $\left| \frac{2}{x} - 3 \right| < 5$.

The given inequality is equivalent to $-5 < \frac{2}{x} - 3 < 5$. Add 3 to obtain $-2 < \frac{2}{x} < 8$, and divide by 2 to get $-1 < \frac{1}{x} < 4$.

Case 1: $x > 0$. Multiply by x to get $-x < 1 < 4x$. Then $x > \frac{1}{4}$ and $x > -1$; these two inequalities are equivalent to the single inequality $x > \frac{1}{4}$.

Case 2: $x < 0$. Multiply by x to obtain $-x > 1 > 4x$. (Note that the inequalities have been reversed, since we multiplied by the negative number x .) Then $x < \frac{1}{4}$ and $x < -1$. These two inequalities are equivalent to $x < -1$.

Thus, the solutions are $x > \frac{1}{4}$ or $x < -1$, the union of the two infinite intervals $(\frac{1}{4}, \infty)$ and $(-\infty, -1)$.

8. Solve $|2x - 5| \geq 3$.

Let us first solve the negation $|2x - 5| < 3$. The latter is equivalent to $-3 < 2x - 5 < 3$. Add 5 to obtain $2 < 2x < 8$, and divide by 2 to obtain $1 < x < 4$. Since this is the solution of the negation, the original inequality has the solution $x \leq 1$ or $x \geq 4$.

9. Solve: $x^2 < 3x + 10$.

$$\begin{aligned} x^2 &< 3x + 10 \\ x^2 - 3x - 10 &< 0 & \text{(Subtract } 3x + 10) \\ (x - 5)(x + 2) &< 0 \end{aligned}$$

The crucial numbers are -2 and 5 . $(x - 5)(x + 2) > 0$ when $x < -2$ (since both $x - 5$ and $x + 2$ are negative); it becomes negative as we pass through -2 (since $x + 2$ changes sign); and then it becomes positive as we pass through 5 (since $x - 5$ changes sign). Thus, the solutions are $-2 < x < 5$.

SUPPLEMENTARY PROBLEMS

10. Describe and diagram the set determined by each of the following conditions:

- | | |
|-----------------------------|---------------------------------|
| (a) $-5 < x < 0$ | (b) $x \leq 0$ |
| (c) $-2 \leq x < 3$ | (d) $x \geq 1$ |
| (e) $ x < 3$ | (f) $ x \geq 5$ |
| (g) $ x - 2 < \frac{1}{2}$ | (h) $ x - 3 > 1$ |
| (i) $0 < x - 2 < 1$ | (j) $0 < x + 3 < \frac{1}{2}$ |
| (k) $ x - 2 \geq 1$ | |

Ans. (e) $-3 < x < 3$; (f) $x \geq 5$ or $x \leq -5$; (g) $\frac{3}{2} < x < \frac{5}{2}$; (h) $x > -2$ or $x < -4$; (i) $x \neq 2$ and $1 < x < 3$; (j) $-\frac{13}{2} < x < -\frac{11}{2}$; (k) $x \geq 3$ or $x \leq 1$

11. Describe and diagram the set determined by each of the following conditions:

- (a) $|3x - 7| < 2$
 (b) $|4x - 11| \geq 1$
 (c) $\left| \frac{x}{3} - 2 \right| \leq 4$

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(d) $\left| \frac{3}{x} - 2 \right| \leq 4$

(e) $\left| 2 + \frac{1}{x} \right| > 1$

(f) $\left| \frac{4}{x} \right| < 3$

Ans. (a) $\frac{5}{3} < x < 3$; (b) $x \geq \frac{1}{2}$ or $x \leq 0$; (c) $-6 \leq x \leq 18$; (d) $x \leq -\frac{3}{2}$ or $x \geq \frac{1}{2}$; (e) $x > 0$ or $x < -1$ or $-\frac{1}{3} < x < 0$;
(f) $x > \frac{4}{3}$ or $x < -\frac{4}{3}$

12. Describe and diagram the set determined by each of the following conditions:

- (a) $x(x-5) < 0$
- (b) $(x-2)(x-6) > 0$
- (c) $(x+1)(x-2) < 0$
- (d) $x(x-2)(x+3) > 0$
- (e) $(x+2)(x+3)(x+4) < 0$
- (f) $(x-1)(x+1)(x-2)(x+3) > 0$
- (g) $(x-1)^2(x+4) > 0$
- (h) $(x-3)(x+5)(x-4)^2 < 0$
- (i) $(x-2)^3 > 0$
- (j) $(x+1)^3 < 0$
- (k) $(x-2)^3(x+1) < 0$
- (l) $(x-1)^3(x+1)^4 < 0$
- (m) $(3x-1)(2x+3) > 0$
- (n) $(x-4)(2x-3) < 0$

Ans. (a) $0 < x < 5$; (b) $x > 6$ or $x < 2$; (c) $-1 < x < 2$; (d) $x > 2$ or $-3 < x < 0$; (e) $-3 < x < -2$ or $x < -4$;
(f) $x > 2$ or $-1 < x < 1$ or $x < -3$; (g) $x > -4$ and $x \neq 1$; (h) $-5 < x < 3$; (i) $x > 2$; (j) $x < -1$;
(k) $-1 < x < 2$; (l) $x < 1$ and $x \neq -1$; (m) $x > \frac{1}{3}$ or $x < -\frac{3}{2}$; (n) $\frac{3}{2} < x < 4$

13. Describe and diagram the set determined by each of the following conditions:

- (a) $x^2 < 4$
- (b) $x^2 \geq 9$
- (c) $(x-2)^2 \leq 16$
- (d) $(2x+1)^2 > 1$
- (e) $x^2 + 3x - 4 > 0$
- (f) $x^2 + 6x + 8 \leq 0$
- (g) $x^2 < 5x + 14$
- (h) $2x^2 > x + 6$
- (i) $6x^2 + 13x < 5$
- (j) $x^3 + 3x^2 > 10x$

Ans. (a) $-2 < x < 2$; (b) $x \geq 3$ or $x \leq -3$; (c) $-2 \leq x \leq 6$; (d) $x > 0$ or $x < -1$; (e) $x > 1$ or $x < -4$; (f) $-4 \leq x \leq -2$;
(g) $-2 < x < 7$; (h) $x > 2$ or $x < -\frac{3}{2}$; (i) $-\frac{5}{2} < x < \frac{1}{3}$; (j) $-5 < x < 0$ or $x > 2$

14. Solve: (a) $-4 < 2 - x < 7$ (b) $\frac{2x-1}{x} < 3$ (c) $\frac{x}{x+2} < 1$
(d) $\frac{3x-1}{2x+3} > 3$ (e) $\left| \frac{2x-1}{x} \right| > 2$ (f) $\left| \frac{x}{x+2} \right| \leq 2$

Ans. (a) $-5 < x < 6$; (b) $x > 0$ or $x < -1$; (c) $x > -2$; (d) $-\frac{10}{3} < x < \frac{3}{2}$; (e) $x < 0$ or $0 < x < \frac{1}{4}$; (f) $x \leq -4$ or $x \geq -1$

15. Solve:

- (a) $|4x - 5| = 3$
- (b) $|x + 6| = 2$
- (c) $|3x - 4| = |2x + 1|$
- (d) $|x + 1| = |x + 2|$
- (e) $|x + 1| = 3x - 1$
- (f) $|x + 1| < |3x - 1|$
- (g) $|3x - 4| \geq |2x + 1|$

Ans. (a) $x = 2$ or $x = \frac{1}{2}$; (b) $x = -4$ or $x = -8$; (c) $x = 5$ or $x = \frac{5}{3}$; (d) $x = -\frac{3}{2}$; (e) $x = 1$; (f) $x > 1$ or $x < 0$;
 (g) $x \geq 5$ or $x \leq \frac{3}{5}$

16. Prove:

- (a) $|x^2| = |x|^2$;
- (b) $|x^n| = |x|^n$ for every integer n ;
- (c) $|x| = \sqrt{x^2}$;
- (d) $|x - y| \leq |x| + |y|$;
- (e) $|x - y| \geq ||x| - |y||$

[Hint: In (e), prove that $|x - y| \geq |x| - |y|$ and $|x - y| \geq |y| - |x|$.]

CHAPTER 2

Rectangular Coordinate Systems

Coordinate Axes

In any plane $\mathcal{P}$, choose a pair of perpendicular lines. Let one of the lines be horizontal. Then the other line must be vertical. The horizontal line is called the x axis, and the vertical line the y axis. (See Fig. 2-1.)

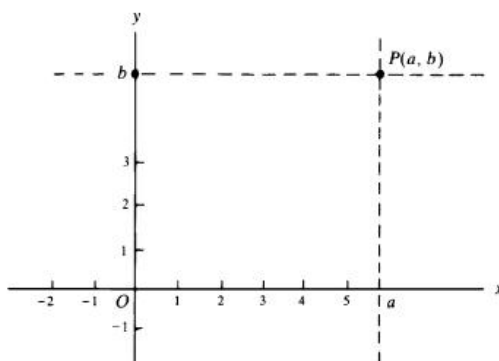


Fig. 2-1

Now choose linear coordinate systems on the x axis and the y axis satisfying the following conditions: The origin for each coordinate system is the point O at which the axes intersect. The x axis is directed from left to right, and the y axis from bottom to top. The part of the x axis with positive coordinates is called the *positive x axis*, and the part of the y axis with positive coordinates is called the *positive y axis*.

We shall establish a correspondence between the points of the plane $\mathcal{P}$ and pairs of real numbers.

Coordinates

Consider any point P of the plane (Fig. 2-1). The vertical line through P intersects the x axis at a unique point; let a be the coordinate of this point on the x axis. The number a is called the x coordinate of P (or the *abscissa* of P). The horizontal line through P intersects the y axis at a unique point; let b be the coordinate of this point on the y axis. The number b is called the y coordinate of P (or the *ordinate* of P). In this way, every point P has a unique pair (a, b) of real numbers associated with it. Conversely, every pair (a, b) of real numbers is associated with a unique point in the plane.

The coordinates of several points are shown in Fig. 2-2. For the sake of simplicity, we have limited them to integers.

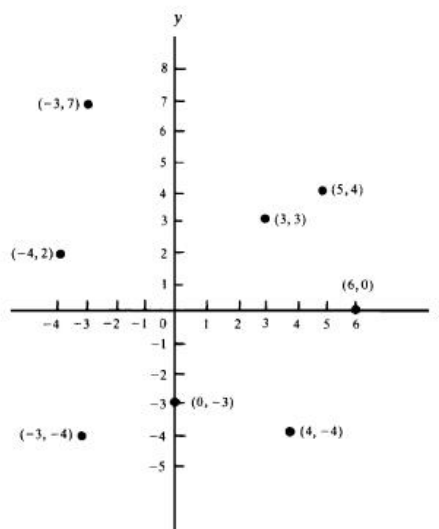


Fig. 2-2

EXAMPLE 2.1: In the coordinate system of Fig. 2-3, to find the point having coordinates $(2, 3)$, start at the origin, move two units to the *right*, and then three units *upward*.

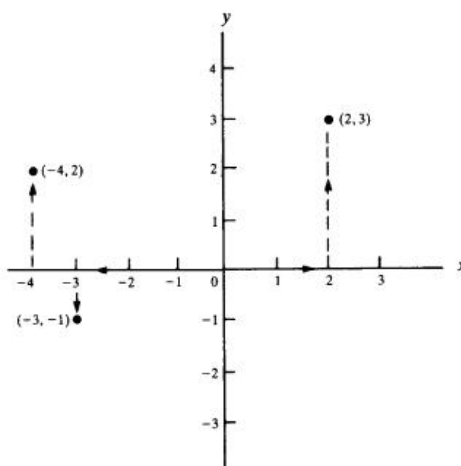


Fig. 2-3

To find the point with coordinates $(-4, 2)$, start at the origin, move four units to the *left*, and then two units *upward*.

To find the point with coordinates $(-3, -1)$, start at the origin, move three units to the *left*, and then one unit *downward*.

The order of these moves is not important. Hence, for example, the point $(2, 3)$ can also be reached by starting at the origin, moving three units *upward*, and then two units to the *right*.

Quadrants

Assume that a coordinate system has been established in the plane $\mathcal{P}$. Then the whole plane $\mathcal{P}$, with the exception of the coordinate axes, can be divided into four equal parts, called *quadrants*. All points with both coordinates positive form the first quadrant, called quadrant I, in the upper-right-hand corner (see Fig. 2-4).

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Quadrant II consists of all points with negative x coordinate and positive y coordinate. *Quadrants III and IV* are also shown in Fig. 2-4.

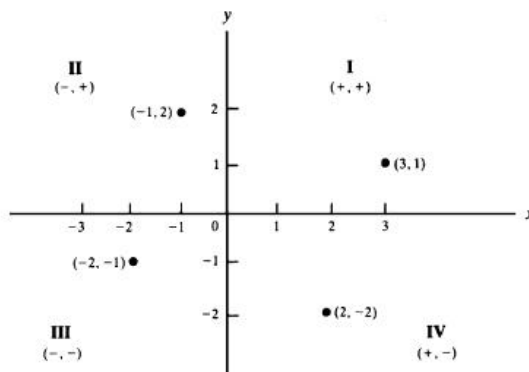


Fig. 2-4

The points on the x axis have coordinates of the form $(a, 0)$. The y axis consists of the points with coordinates of the form $(0, b)$.

Given a coordinate system, it is customary to refer to the point with coordinates (a, b) as “the point (a, b) .” For example, one might say, “The point $(0, 1)$ lies on the y axis.”

The Distance Formula

The distance $\overline{P_1P_2}$ between points P_1 and P_2 with coordinates (x_1, y_1) and (x_2, y_2) in a given coordinate system (see Fig. 2-5) is given by the following distance formula:

$$\overline{P_1P_2} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (2.1)$$

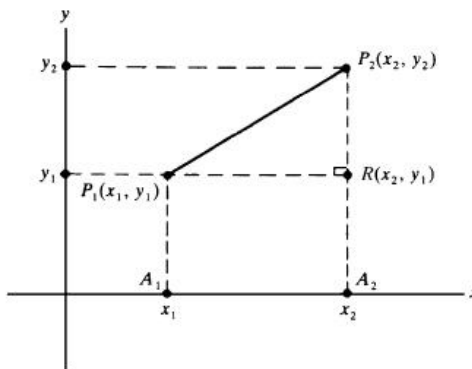


Fig. 2-5

To see this, let R be the point where the vertical line through P_2 intersects the horizontal line through P_1 . The x coordinate of R is x_2 , the same as that of P_2 . The y coordinate of R is y_1 , the same as that of P_1 . By the Pythagorean theorem, $(\overline{P_1P_2})^2 = (\overline{P_1R})^2 + (\overline{P_2R})^2$. If A_1 and A_2 are the projections of P_1 and P_2 on the x axis, the segments P_1R and A_1A_2 are opposite sides of a rectangle, so that $\overline{P_1R} = \overline{A_1A_2}$. But $\overline{A_1A_2} = |x_1 - x_2|$ by property (1.12). So, $\overline{P_1R} = |x_1 - x_2|$. Similarly, $\overline{P_2R} = |y_1 - y_2|$. Hence, $(\overline{P_1P_2})^2 = |x_1 - x_2|^2 + |y_1 - y_2|^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$.

Taking square roots, we obtain the distance formula. (It can be checked that the formula also is valid when P_1 and P_2 lie on the same vertical or horizontal line.)

EXAMPLES:

- (a) The distance between $(2, 5)$ and $(7, 17)$ is

$$\sqrt{(2-7)^2 + (5-17)^2} = \sqrt{(-5)^2 + (-12)^2} = \sqrt{25+144} = \sqrt{169} = 13$$

- (b) The distance between $(1, 4)$ and $(5, 2)$ is

$$\sqrt{(1-5)^2 + (4-2)^2} = \sqrt{(-4)^2 + (2)^2} = \sqrt{16+4} = \sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$$

The Midpoint Formulas

The point $M(x, y)$ that is the midpoint of the segment connecting the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ has the coordinates

$$x = \frac{x_1 + x_2}{2} \quad y = \frac{y_1 + y_2}{2} \quad (2.2)$$

Thus, the coordinates of the midpoints are the averages of the coordinates of the endpoints. See Fig. 2-6.

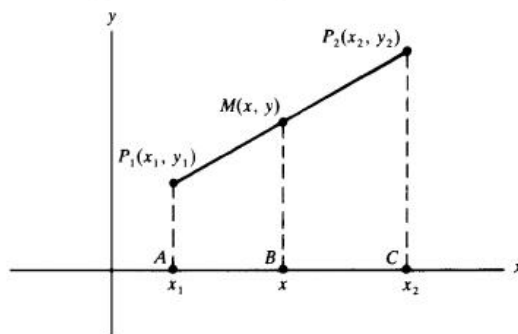


Fig. 2-6

To see this, let A, B, C be the projections of P_1, M, P_2 on the x axis. The x coordinates of A, B, C are x_1, x, x_2 . Since the lines P_1A, MB , and P_2C are parallel, the ratios $\overline{P_1M}/\overline{MP_2}$ and $\overline{AB}/\overline{BC}$ are equal. Since $\overline{P_1M} = \overline{MP_2}$, $\overline{AB} = \overline{BC}$. Since $\overline{AB} = x - x_1$ and $\overline{BC} = x_2 - x$,

$$x - x_1 = x_2 - x$$

$$2x = x_1 + x_2$$

$$x = \frac{x_1 + x_2}{2}$$

(The same equation holds when P_2 is to the left of P_1 , in which case $\overline{AB} = x_1 - x$ and $\overline{BC} = x - x_2$.) Similarly, $y = (y_1 + y_2)/2$.

EXAMPLES:

- (a) The midpoint of the segment connecting $(2, 9)$ and $(4, 3)$ is $\left(\frac{2+4}{2}, \frac{9+3}{2}\right) = (3, 6)$.

- (b) The point halfway between $(-5, 1)$ and $(1, 4)$ is $\left(\frac{-5+1}{2}, \frac{1+4}{2}\right) = \left(-2, \frac{5}{2}\right)$.

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Proofs of Geometric Theorems

Proofs of geometric theorems can often be given more easily by use of coordinates than by deductions from axioms and previously derived theorems. Proofs by means of coordinates are called *analytic*, in contrast to so-called *synthetic* proofs from axioms.

EXAMPLE 2.2: Let us prove analytically that the segment joining the midpoints of two sides of a triangle is one-half the length of the third side. Construct a coordinate system so that the third side AB lies on the positive x axis, A is the origin, and the third vertex C lies above the x axis, as in Fig. 2-7.

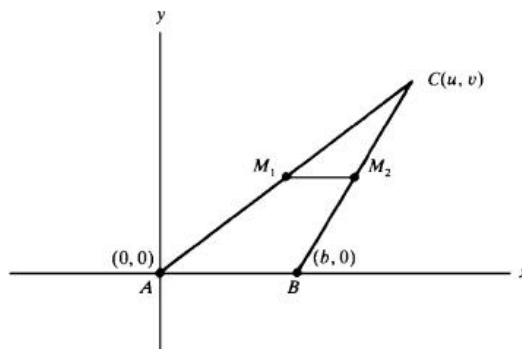


Fig. 2-7

Let b be the x coordinate of B . (In other words, let $b = \overline{AB}$.) Let C have coordinates (u, v) . Let M_1 and M_2 be the midpoints of sides AC and BC , respectively. By the midpoint formulas (2.2), the coordinates of M_1 are $\left(\frac{u}{2}, \frac{v}{2}\right)$, and the coordinates of M_2 are $\left(\frac{u+b}{2}, \frac{v}{2}\right)$. By the distance formula (2.1),

$$\overline{M_1M_2} = \sqrt{\left(\frac{u}{2} - \frac{u+b}{2}\right)^2 + \left(\frac{v}{2} - \frac{v}{2}\right)^2} = \sqrt{\left(\frac{b}{2}\right)^2} = \frac{b}{2}$$

which is half the length of side AB .

SOLVED PROBLEMS

- Show that the distance between a point $P(x, y)$ and the origin is $\sqrt{x^2 + y^2}$.
Since the origin has coordinates $(0, 0)$, the distance formula yields $\sqrt{(x-0)^2 + (y-0)^2} = \sqrt{x^2 + y^2}$.
- Is the triangle with vertices $A(1, 5)$, $B(4, 2)$, and $C(5, 6)$ isosceles?

$$\overline{AB} = \sqrt{(1-4)^2 + (5-2)^2} = \sqrt{(-3)^2 + (3)^2} = \sqrt{9+9} = \sqrt{18}$$

$$\overline{AC} = \sqrt{(1-5)^2 + (5-6)^2} = \sqrt{(-4)^2 + (-1)^2} = \sqrt{16+1} = \sqrt{17}$$

$$\overline{BC} = \sqrt{(4-5)^2 + (2-6)^2} = \sqrt{(-1)^2 + (-4)^2} = \sqrt{1+16} = \sqrt{17}$$

Since $\overline{AC} = \overline{BC}$, the triangle is isosceles.
- Is the triangle with vertices $A(-5, 6)$, $B(2, 3)$, and $C(5, 10)$ a right triangle?

$$\overline{AB} = \sqrt{(-5-2)^2 + (6-3)^2} = \sqrt{(-7)^2 + (3)^2} = \sqrt{49+9} = \sqrt{58}$$

$$\begin{aligned}\overline{AC} &= \sqrt{(-5-5)^2 + (6-10)^2} = \sqrt{(-10)^2 + (-4)^2} \\ &= \sqrt{100+16} = \sqrt{116}\end{aligned}$$

$$\overline{BC} = \sqrt{(2-5)^2 + (3-10)^2} = \sqrt{(-3)^2 + (-7)^2} = \sqrt{9+49} = \sqrt{58}$$

Since $\overline{AC}^2 = \overline{AB}^2 + \overline{BC}^2$, the converse of the Pythagorean theorem tells us that $\triangle ABC$ is a right triangle, with right angle at B ; in fact, since $\overline{AB} = \overline{BC}$, $\triangle ABC$ is an isosceles right triangle.

4. Prove analytically that, if the medians to two sides of a triangle are equal, then those sides are equal. (Recall that a *median* of a triangle is a line segment joining a vertex to the midpoint of the opposite side.)

In $\triangle ABC$, let M_1 and M_2 be the midpoints of sides AC and BC , respectively. Construct a coordinate system so that A is the origin, B lies on the positive x axis, and C lies above the x axis (see Fig. 2-8). Assume that $\overline{AM_2} = \overline{BM_1}$. We must prove that $\overline{AC} = \overline{BC}$. Let b be the x coordinate of B , and let C have coordinates (u, v) . Then, by the midpoint formulas, M_1 has coordinates $\left(\frac{u}{2}, \frac{v}{2}\right)$, and M_2 has coordinates $\left(\frac{u+b}{2}, \frac{v}{2}\right)$. Hence,

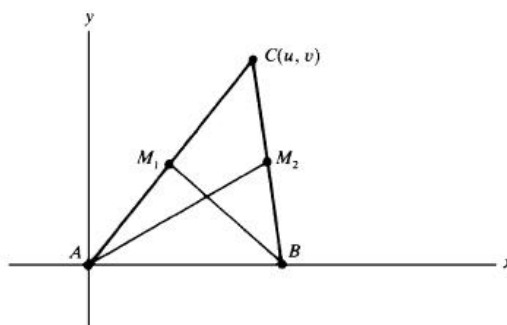


Fig. 2-8

$$\overline{AM_2} = \sqrt{\left(\frac{u+b}{2}\right)^2 + \left(\frac{v}{2}\right)^2} \quad \text{and} \quad \overline{BM_1} = \sqrt{\left(\frac{u}{2}-b\right)^2 + \left(\frac{v}{2}\right)^2}$$

Since $\overline{AM_2} = \overline{BM_1}$,

$$\left(\frac{u+b}{2}\right)^2 + \left(\frac{v}{2}\right)^2 = \left(\frac{u}{2}-b\right)^2 + \left(\frac{v}{2}\right)^2 = \left(\frac{u-2b}{2}\right)^2 + \left(\frac{v}{2}\right)^2$$

Hence, $\frac{(u+b)^2}{4} + \frac{v^2}{4} = \frac{(u-2b)^2}{4} + \frac{v^2}{4}$ and, therefore, $(u+b)^2 = (u-2b)^2$. So, $u+b = \pm(u-2b)$. If $u+b = u-2b$, then $b = -2b$, and therefore, $b = 0$, which is impossible, since $A \neq B$. Hence, $u+b = -(u-2b) = -u+2b$, whence $2u = b$. Now $\overline{BC} = \sqrt{(u-b)^2 + v^2} = \sqrt{(u-2u)^2 + v^2} = \sqrt{(-u)^2 + v^2} = \sqrt{u^2 + v^2}$, and $\overline{AC} = \sqrt{u^2 + v^2}$. Thus, $\overline{AC} = \overline{BC}$.

5. Find the coordinates (x, y) of the point Q on the line segment joining $P_1(1, 2)$ and $P_2(6, 7)$, such that Q divides the segment in the ratio 2:3, that is, such that $\overline{P_1Q}/\overline{QP_2} = \frac{2}{3}$.

Let the projections of P_1 , Q , and P_2 on the x axis be A_1 , Q' , and A_2 , with x coordinates 1, x , and 6, respectively (see Fig. 2-9). Now $\overline{A_1Q'}/\overline{Q'A_2} = \overline{P_1Q}/\overline{QP_2} = \frac{2}{3}$. (When two lines are cut by three parallel lines, corresponding

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segments are in proportion.) But $\overline{A_1Q'} = x - 1$, and $\overline{Q'A_2} = 6 - x$. So $\frac{x-1}{6-x} = \frac{2}{3}$, and cross-multiplying yields $3x - 3 = 12 - 2x$. Hence $5x = 15$, whence $x = 3$. By similar reasoning, $\frac{y-2}{7-y} = \frac{2}{3}$, from which it follows that $y = 4$.

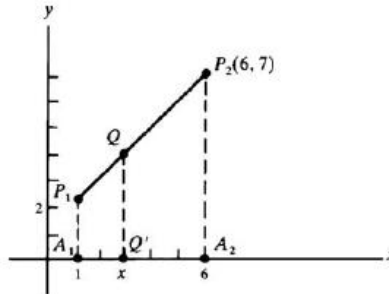


Fig. 2-9

SUPPLEMENTARY PROBLEMS

6. In Fig. 2-10, find the coordinates of points A, B, C, D, E, and F.

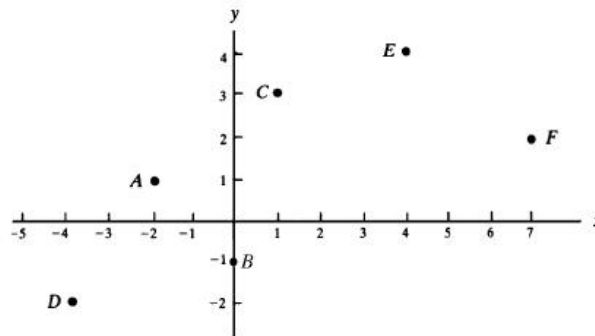


Fig. 2-10

Ans. (A) = (-2, 1); (B) = (0, -1); (C) = (1, 3); (D) = (-4, -2); (E) = (4, 4); (F) = (7, 2).

7. Draw a coordinate system and show the points having the following coordinates: (2, -3), (3, 3), (-1, 1), (2, -2), (0, 3), (3, 0), (-2, 3).

8. Find the distances between the following pairs of points:

- | | | |
|-----------------------|------------------------|-------------------------------------|
| (a) (3, 4) and (3, 6) | (b) (2, 5) and (2, -2) | (c) (3, 1) and (2, 1) |
| (d) (2, 3) and (5, 7) | (e) (-2, 4) and (3, 0) | (f) $(-2, \frac{1}{2})$ and (4, -1) |

Ans. (a) 2; (b) 7; (c) 1; (d) 5; (e) $\sqrt{41}$; (f) $\frac{5}{2}\sqrt{17}$

9. Draw the triangle with vertices A(2, 5), B(2, -5), and C(-3, 5), and find its area.

Ans. Area = 25

10. If $(2, 2)$, $(2, -4)$, and $(5, 2)$ are three vertices of a rectangle, find the fourth vertex.

Ans. $(5, -4)$

11. If the points $(2, 4)$ and $(-1, 3)$ are the opposite vertices of a rectangle whose sides are parallel to the coordinate axes (that is, the x and y axes), find the other two vertices.

Ans. $(-1, 4)$ and $(2, 3)$

12. Determine whether the following triples of points are the vertices of an isosceles triangle: (a) $(4, 3)$, $(1, 4)$, $(3, 10)$; (b) $(-1, 1)$, $(3, 3)$, $(1, -1)$; (c) $(2, 4)$, $(5, 2)$, $(6, 5)$.

Ans. (a) no; (b) yes; (c) no

13. Determine whether the following triples of points are the vertices of a right triangle. For those that are, find the area of the right triangle: (a) $(10, 6)$, $(3, 3)$, $(6, -4)$; (b) $(3, 1)$, $(1, -2)$, $(-3, -1)$; (c) $(5, -2)$, $(0, 3)$, $(2, 4)$.

Ans. (a) yes, area = 29; (b) no; (c) yes, area = $\frac{15}{2}$

14. Find the perimeter of the triangle with vertices $A(4, 9)$, $B(-3, 2)$, and $C(8, -5)$.

Ans. $7\sqrt{2} + \sqrt{170} + 2\sqrt{53}$

15. Find the value or values of y for which $(6, y)$ is equidistant from $(4, 2)$ and $(9, 7)$.

Ans. 5

16. Find the midpoints of the line segments with the following endpoints: (a) $(2, -3)$ and $(7, 4)$; (b) $(\frac{5}{3}, 2)$ and $(4, 1)$; (c) $(\sqrt{3}, 0)$ and $(1, 4)$.

Ans. (a) $(\frac{9}{2}, \frac{1}{2})$; (b) $(\frac{17}{6}, \frac{3}{2})$; (c) $(\frac{1+\sqrt{3}}{2}, 2)$

17. Find the point (x, y) such that $(2, 4)$ is the midpoint of the line segment connecting (x, y) and $(1, 5)$.

Ans. $(3, 3)$

18. Determine the point that is equidistant from the points $A(-1, 7)$, $B(6, 6)$, and $C(5, -1)$.

Ans. $(\frac{52}{25}, \frac{153}{50})$

19. Prove analytically that the midpoint of the hypotenuse of a right triangle is equidistant from the three vertices.

20. Show analytically that the sum of the squares of the distance of any point P from two opposite vertices of a rectangle is equal to the sum of the squares of its distances from the other two vertices.

21. Prove analytically that the sum of the squares of the four sides of a parallelogram is equal to the sum of the squares of the diagonals.

CHAPTER 2 Rectangular Coordinate Systems

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22. Prove analytically that the sum of the squares of the medians of a triangle is equal to three-fourths the sum of the squares of the sides.
23. Prove analytically that the line segments joining the midpoints of opposite sides of a quadrilateral bisect each other.
24. Prove that the coordinates (x, y) of the point Q that divides the line segments from $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ in the ratio $r_1:r_2$ are determined by the formulas

$$x = \frac{r_1x_2 + r_2x_1}{r_1 + r_2} \quad \text{and} \quad y = \frac{r_1y_2 + r_2y_1}{r_1 + r_2}$$

(Hint: Use the reasoning of Problem 5.)

25. Find the coordinates of the point Q on the segment P_1P_2 such that $\overline{P_1Q}/\overline{QP_2} = \frac{2}{7}$, if (a) $P_1 = (0, 0)$, $P_2 = (7, 9)$; (b) $P_1 = (-1, 0)$, $P_2 = (0, 7)$; (c) $P_1 = (-7, -2)$, $P_2 = (2, 7)$; (d) $P_1 = (1, 3)$, $P_2 = (4, 2)$.

Ans. (a) $(\frac{14}{9}, 2)$; (b) $(-\frac{7}{9}, \frac{14}{9})$; (c) $(-5, \frac{28}{9})$; (d) $(\frac{13}{9}, \frac{32}{9})$

CHAPTER 3

Lines

The Steepness of a Line

The steepness of a line is measured by a number called the *slope* of the line. Let $\mathcal{L}$ be any line, and let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be two points of $\mathcal{L}$. The slope of $\mathcal{L}$ is defined to be the number $m = \frac{y_2 - y_1}{x_2 - x_1}$. The slope is the ratio of a change in the y coordinate to the corresponding change in the x coordinate. (See Fig. 3-1.)

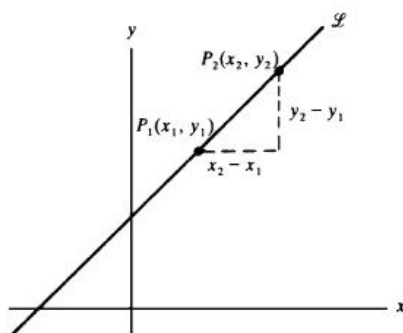


Fig. 3-1

For the definition of the slope to make sense, it is necessary to check that the number m is independent of the choice of the points P_1 and P_2 . If we choose another pair $P_3(x_3, y_3)$ and $P_4(x_4, y_4)$, the same value of m must result. In Fig. 3-2, triangle P_3P_4T is similar to triangle P_1P_2Q . Hence,

$$\frac{\overline{QP_2}}{\overline{P_1Q}} = \frac{\overline{TP_4}}{\overline{P_3T}} \quad \text{or} \quad \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_4 - y_3}{x_4 - x_3}$$

Therefore, P_1 and P_2 determine the same slope as P_3 and P_4 .

EXAMPLE 3.1: The slope of the line joining the points $(1, 2)$ and $(4, 6)$ in Fig. 3-3 is $\frac{6-2}{4-1} = \frac{4}{3}$. Hence, as a point on the line moves 3 units to the right, it moves 4 units upwards. Moreover, the slope is not affected by the order in which the points are given: $\frac{2-6}{1-4} = \frac{-4}{-3} = \frac{4}{3}$. In general, $\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$.

The Sign of the Slope

The sign of the slope has significance. Consider, for example, a line $\mathcal{L}$ that moves upward as it moves to the right, as in Fig. 3-4(a). Since $y_2 > y_1$ and $x_2 > x_1$, we have $m = \frac{y_2 - y_1}{x_2 - x_1} > 0$. The slope of $\mathcal{L}$ is positive.

Now consider a line $\mathcal{L}$ that moves downward as it moves to the right, as in Fig. 3-4(b). Here $y_2 < y_1$ while $x_2 > x_1$; hence, $m = \frac{y_2 - y_1}{x_2 - x_1} < 0$. The slope of $\mathcal{L}$ is negative.

CHAPTER 3 Lines

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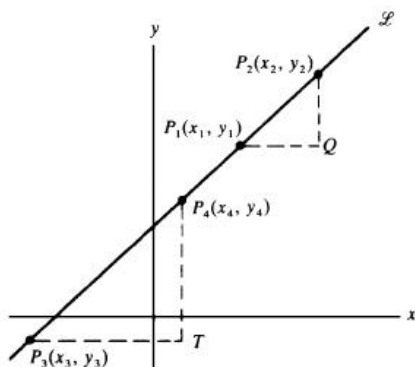


Fig. 3-2

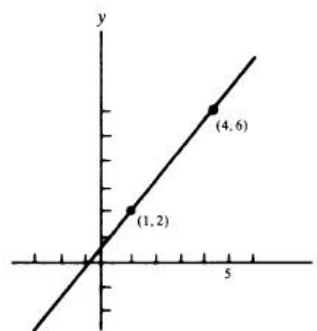


Fig. 3-3

Now let the line $\mathcal{L}$ be horizontal, as in Fig. 3-4(c). Here $y_1 = y_2$, so that $y_2 - y_1 = 0$. In addition, $x_2 - x_1 \neq 0$. Hence, $m = \frac{0}{x_2 - x_1} = 0$. The slope of $\mathcal{L}$ is zero.

Line $\mathcal{L}$ is vertical in Fig. 3-4(d), where we see that $y_2 - y_1 > 0$ while $x_2 - x_1 = 0$. Thus, the expression $\frac{y_2 - y_1}{x_2 - x_1}$ is undefined. The slope is not defined for a vertical line $\mathcal{L}$. (Sometimes we describe this situation by saying that the slope of $\mathcal{L}$ is “infinite.”)

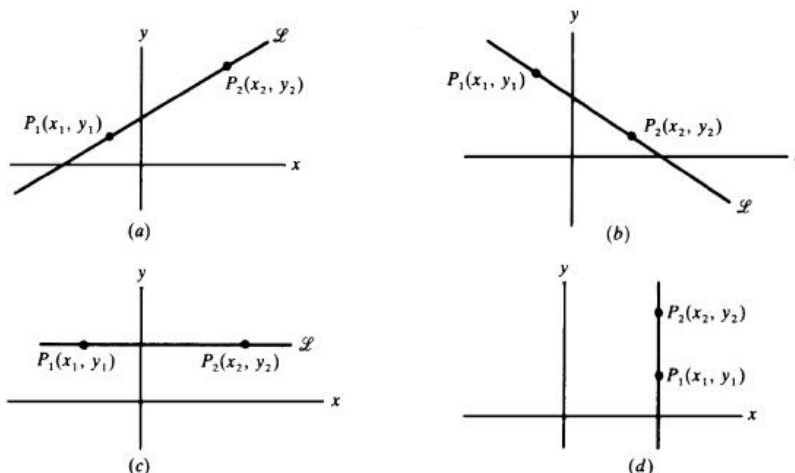


Fig. 3-4

Slope and Steepness

Consider any line $\mathcal{L}$ with positive slope, passing through a point $P_1(x_1, y_1)$; such a line is shown in Fig. 3-5. Choose the point $P_2(x_2, y_2)$ on $\mathcal{L}$ such that $x_2 - x_1 = 1$. Then the slope m of $\mathcal{L}$ is equal to the distance $\overline{AP_2}$. As the steepness of the line increases, $\overline{AP_2}$ increases without limit, as shown in Fig. 3-6(a). Thus, the slope of $\mathcal{L}$ increases without bound from 0 (when $\mathcal{L}$ is horizontal) to $+\infty$ (when the line is vertical). By a similar argument, using Fig. 3-6(b), we can show that as a negatively sloped line becomes steeper, the slope steadily decreases from 0 (when the line is horizontal) to $-\infty$ (when the line is vertical).

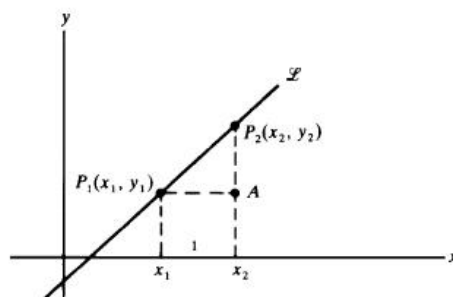


Fig. 3-5

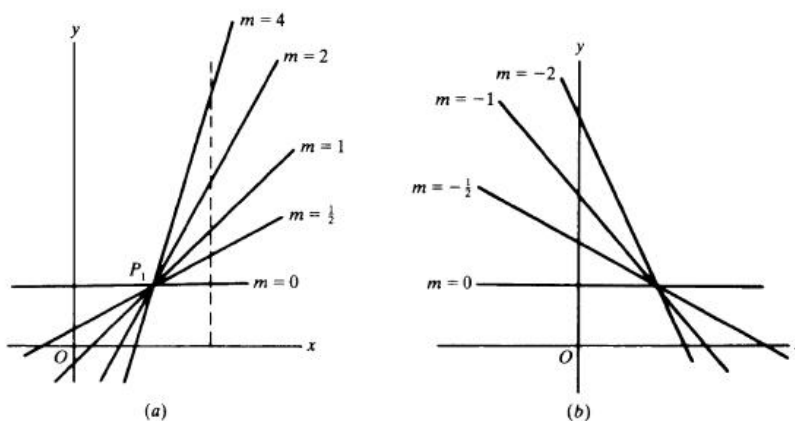


Fig. 3-6

Equations of Lines

Let $\mathcal{L}$ be a line that passes through a point $P_1(x_1, y_1)$ and has slope m , as in Fig. 3-7(a). For any other point $P(x, y)$ on the line, the slope m is, by definition, the ratio of $y - y_1$ to $x - x_1$. Thus, for any point (x, y) on $\mathcal{L}$,

$$m = \frac{y - y_1}{x - x_1} \quad (3.1)$$

Conversely, if $P(x, y)$ is *not* on line $\mathcal{L}$ as in Fig. 3-7(b), then the slope $\frac{y - y_1}{x - x_1}$ of the line PP_1 is different from the slope m of $\mathcal{L}$; hence (3.1) does not hold for points that are not on $\mathcal{L}$. Thus, the line consists of only those points (x, y) that satisfy (3.1). In such a case, we say that $\mathcal{L}$ is the *graph* of (3.1).

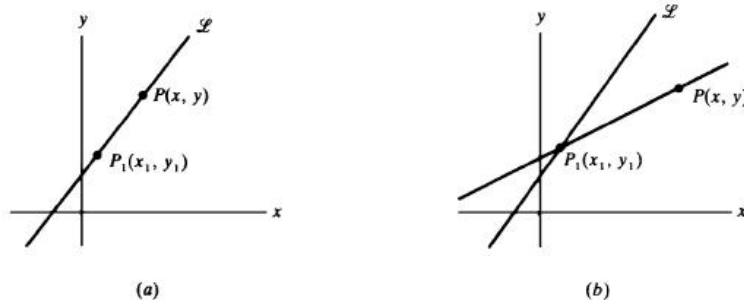


Fig. 3-7

CHAPTER 3 Lines

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A Point-Slope Equation

A point-slope equation of the line $\mathcal{L}$ is any equation of the form (3.1). If the slope m of $\mathcal{L}$ is known, then each point (x_1, y_1) of $\mathcal{L}$ yields a point-slope equation of $\mathcal{L}$. Hence, there is an infinite number of point-slope equations for $\mathcal{L}$. Equation (3.1) is equivalent to $y - y_1 = m(x - x_1)$.

EXAMPLE 3.2: (a) The line passing through the point $(2, 5)$ with slope 3 has a point-slope equation $\frac{y-5}{x-2} = 3$.
 (b) Let $\mathcal{L}$ be the line through the points $(3, -1)$ and $(2, 3)$. Its slope is $m = \frac{3-(-1)}{2-3} = \frac{4}{-1} = -4$. Two point-slope equations of $\mathcal{L}$ are $\frac{y+1}{x-3} = -4$ and $\frac{y-3}{x-2} = -4$.

Slope-Intercept Equation

If we multiply (3.1) by $x - x_1$, we obtain the equation $y - y_1 = m(x - x_1)$, which can be reduced first to $y - y_1 = mx - mx_1$, and then to $y = mx + (y_1 - mx_1)$. Let b stand for the number $y_1 - mx_1$. Then the equation for line $\mathcal{L}$ becomes

$$y = mx + b \quad (3.2)$$

Equation (3.2) yields the value $y = b$ when $x = 0$, so the point $(0, b)$ lies on $\mathcal{L}$. Thus, b is the y coordinate of the intersection of $\mathcal{L}$ and the y axis, as shown in Fig. 3-8. The number b is called the y intercept of $\mathcal{L}$, and (3.2) is called the *slope-intercept equation* for $\mathcal{L}$.

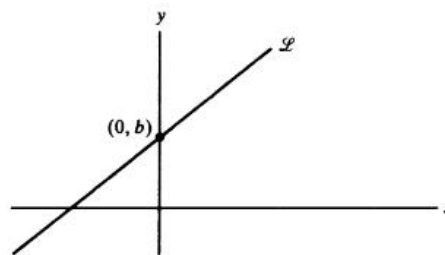


Fig. 3-8

EXAMPLE 3.3: The line through the points $(2, 3)$ and $(4, 9)$ has slope

$$m = \frac{9-3}{4-2} = \frac{6}{2} = 3$$

Its slope-intercept equation has the form $y = 3x + b$. Since the point $(2, 3)$ lies on the line, $(2, 3)$ must satisfy this equation. Substitution yields $3 = 3(2) + b$, from which we find $b = -3$. Thus, the slope-intercept equation is $y = 3x - 3$.

Another method for finding this equation is to write a point-slope equation of the line, say $\frac{y-3}{x-2} = 3$. Then multiplying by $x - 2$ and adding 3 yields $y = 3x - 3$.

Parallel Lines

Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be parallel nonvertical lines, and let A_1 and A_2 be the points at which $\mathcal{L}_1$ and $\mathcal{L}_2$ intersect the y axis, as in Fig. 3-9(a). Further, let B_1 be one unit to the right of A_1 , and B_2 one unit to the right of A_2 . Let C_1 and C_2 be the intersections of the verticals through B_1 and B_2 with $\mathcal{L}_1$ and $\mathcal{L}_2$. Now, triangle $A_1B_1C_1$ is congruent to triangle $A_2B_2C_2$ (by the angle-side-angle congruence theorem). Hence, $\overline{B_1C_1} = \overline{B_2C_2}$ and

$$\text{Slope of } \mathcal{L}_1 = \frac{\overline{B_1C_1}}{1} = \frac{\overline{B_2C_2}}{1} = \text{slope of } \mathcal{L}_2$$

Thus, parallel lines have equal slopes.

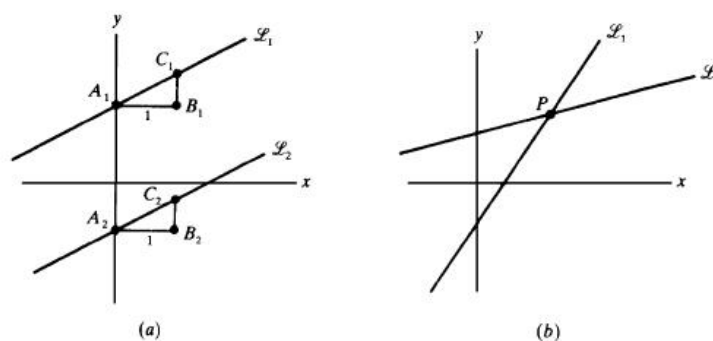


Fig. 3-9

Conversely, assume that two different lines $\mathcal{L}_1$ and $\mathcal{L}_2$ are not parallel, and let them meet at point P , as in Fig. 3-9(b). If $\mathcal{L}_1$ and $\mathcal{L}_2$ had the same slope, then they would have to be the same line. Hence, $\mathcal{L}_1$ and $\mathcal{L}_2$ have different slopes.

Theorem 3.1: Two distinct nonvertical lines are parallel if and only if their slopes are equal.

EXAMPLE 3.4: Find the slope-intercept equation of the line $\mathcal{L}$ through $(4, 1)$ and parallel to the line $\mathcal{M}$ having the equation $4x - 2y = 5$.

By solving the latter equation for y , we see that $\mathcal{M}$ has the slope-intercept equation $y = 2x - \frac{5}{2}$. Hence, $\mathcal{M}$ has slope 2. The slope of the parallel line $\mathcal{L}$ also must be 2. So the slope-intercept equation of $\mathcal{L}$ has the form $y = 2x + b$. Since $(4, 1)$ lies on $\mathcal{L}$, we can write $1 = 2(4) + b$. Hence, $b = -7$, and the slope-intercept equation of $\mathcal{L}$ is $y = 2x - 7$.

Perpendicular Lines

In Problem 5 we shall prove the following:

Theorem 3.2: Two nonvertical lines are perpendicular if and only if the product of their slopes is -1 .

If m_1 and m_2 are the slopes of perpendicular lines, then $m_1 m_2 = -1$. This is equivalent to $m_2 = -\frac{1}{m_1}$; hence, the slopes of perpendicular lines are negative reciprocals of each other.

SOLVED PROBLEMS

- Find the slope of the line having the equation $3x - 4y = 8$. Draw the line. Do the points $(6, 2)$ and $(12, 7)$ lie on the line?

Solving the equation for y yields $y = \frac{3}{4}x - 2$. This is the slope-intercept equation; the slope is $\frac{3}{4}$ and the y intercept is -2 .

Substituting 0 for x shows that the line passes through the point $(0, -2)$. To draw the line, we need another point. If we substitute 4 for x in the slope-intercept equation, we get $y = \frac{3}{4}(4) - 2 = 1$. So, $(4, 1)$ also lies on the line, which is drawn in Fig. 3-10. (We could have found other points on the line by substituting numbers other than 4 for x .)

To test whether $(6, 2)$ is on the line, we substitute 6 for x and 2 for y in the original equation, $3x - 4y = 8$. The two sides turn out to be unequal; hence, $(6, 2)$ is not on the line. The same procedure shows that $(12, 7)$ lies on the line.

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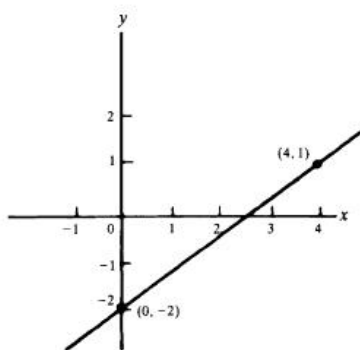


Fig. 3-10

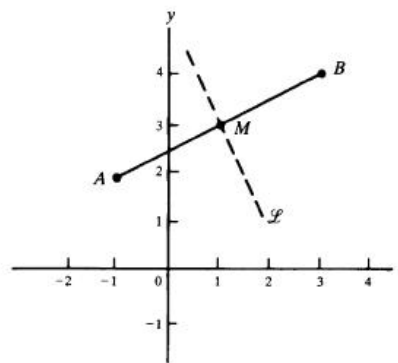


Fig. 3-11

2. Let $\mathcal{L}$ be the perpendicular bisector of the line segment joining the points $A(-1, 2)$ and $B(3, 4)$, as shown in Fig. 3-11. Find an equation for $\mathcal{L}$.

$\mathcal{L}$ passes through the midpoint M of segment AB . By the midpoint formulas (2.2), the coordinates of M are $(1, 3)$. The slope of the line through A and B is $\frac{4-2}{3-(-1)} = \frac{2}{4} = \frac{1}{2}$. Let m be the slope of $\mathcal{L}$. By Theorem 3.2, $\frac{1}{2}m = -1$, whence $m = -2$.

The slope-intercept equation for $\mathcal{L}$ has the form $y = -2x + b$. Since $M(1, 3)$ lies on $\mathcal{L}$, we have $3 = -2(1) + b$. Hence, $b = 5$, and the slope-intercept equation of $\mathcal{L}$ is $y = -2x + 5$.

3. Determine whether the points $A(1, -1)$, $B(3, 2)$, and $C(7, 8)$ are collinear, that is, lie on the same line.

A , B , and C are collinear if and only if the line AB is identical with the line AC , which is equivalent to the slope of AB being equal to the slope of AC . The slopes of AB and AC are $\frac{2-(-1)}{3-1} = \frac{3}{2}$ and $\frac{8-(-1)}{7-1} = \frac{9}{6} = \frac{3}{2}$. Hence, A , B , and C are collinear.

4. Prove analytically that the figure obtained by joining the midpoints of consecutive sides of a quadrilateral is a parallelogram.

Locate a quadrilateral with consecutive vertices, A , B , C , and D on a coordinate system so that A is the origin, B lies on the positive x axis, and C and D lie above the x axis. (See Fig. 3-12.) Let b be the x coordinate of B , (u, v) the coordinates of C , and (x, y) the coordinates of D . Then, by the midpoint formula (2.2), the midpoints M_1 , M_2 , M_3 , and M_4 of sides AB , BC , CD , and DA have coordinates $(\frac{b}{2}, 0)$, $(\frac{u+b}{2}, \frac{v}{2})$, $(\frac{x+u}{2}, \frac{y+v}{2})$, and $(\frac{x}{2}, \frac{y}{2})$, respectively. We must show that $M_1M_2M_3M_4$ is a parallelogram. To do this, it suffices to prove that lines M_1M_2 and M_3M_4 are parallel and that lines M_2M_3 and M_1M_4 are parallel. Let us calculate the slopes of these lines:

$$\text{slope}(M_1M_2) = \frac{\frac{v}{2} - 0}{\frac{u+b}{2} - \frac{b}{2}} = \frac{\frac{v}{2}}{\frac{u}{2}} = \frac{v}{u}$$

$$\text{slope}(M_3M_4) = \frac{\frac{y}{2} - \frac{y+v}{2}}{\frac{x}{2} - \frac{x+u}{2}} = \frac{-\frac{v}{2}}{-\frac{u}{2}} = \frac{v}{u}$$

$$\text{slope}(M_2M_3) = \frac{\frac{y+v}{2} - \frac{v}{2}}{\frac{x+u}{2} - \frac{u+b}{2}} = \frac{\frac{y}{2}}{\frac{x-b}{2}} = \frac{y}{x-b}$$

$$\text{slope}(M_1M_4) = \frac{\frac{y}{2} - 0}{\frac{x}{2} - \frac{b}{2}} = \frac{y}{x-b}$$

Since $\text{slope}(M_1M_2) = \text{slope}(M_3M_4)$, M_1M_2 and M_3M_4 are parallel. Since $\text{slope}(M_2M_3) = \text{slope}(M_1M_4)$, M_2M_3 and M_1M_4 are parallel. Thus, $M_1M_2M_3M_4$ is a parallelogram.

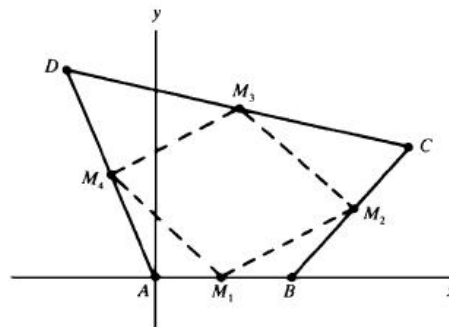


Fig. 3-12

5. Prove Theorem 3.2.

First we assume $\mathcal{L}_1$ and $\mathcal{L}_2$ are perpendicular nonvertical lines with slopes m_1 and m_2 . We must show that $m_1 m_2 = -1$. Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be the lines through the origin O that are parallel to $\mathcal{L}_1$ and $\mathcal{L}_2$, as shown in Fig. 3-13(a). Then the slope of $\mathcal{M}_1$ is m_1 , and the slope of $\mathcal{M}_2$ is m_2 (by Theorem 3.1). Moreover, $\mathcal{M}_1$ and $\mathcal{M}_2$ are perpendicular, since $\mathcal{L}_1$ and $\mathcal{L}_2$ are perpendicular.

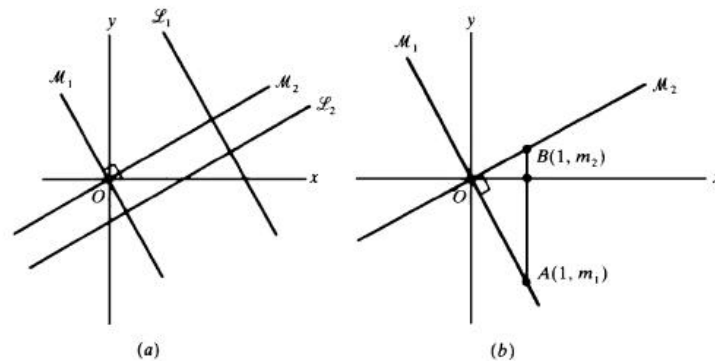


Fig. 3-13

Now let A be the point on $\mathcal{M}_1$ with x coordinate 1, and let B be the point on $\mathcal{M}_2$ with x coordinate 1, as in Fig. 3-13(b). The slope-intercept equation of $\mathcal{M}_1$ is $y = m_1 x$; therefore, the y coordinate of A is m_1 , since its x coordinate is 1. Similarly, the y coordinate of B is m_2 . By the distance formula (2.1),

$$\overline{OB} = \sqrt{(1-0)^2 + (m_2-0)^2} = \sqrt{1+m_2^2}$$

$$\overline{OA} = \sqrt{(1-0)^2 + (m_1-0)^2} = \sqrt{1+m_1^2}$$

$$\overline{BA} = \sqrt{(1-1)^2 + (m_2-m_1)^2} = \sqrt{(m_2-m_1)^2}$$

Then by the Pythagorean theorem for right triangle BOA ,

$$\overline{BA}^2 = \overline{OB}^2 + \overline{OA}^2$$

or

$$(m_2 - m_1)^2 = (1 + m_2^2) + (1 + m_1^2)$$

$$m_2^2 - 2m_2m_1 + m_1^2 = 2 + m_2^2 + m_1^2$$

$$m_2m_1 = -1$$

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Now, conversely, we assume that $m_1 m_2 = -1$, where m_1 and m_2 are the slopes of nonvertical lines $\mathcal{L}_1$ and $\mathcal{L}_2$. Then $\mathcal{L}_1$ is not parallel to $\mathcal{L}_2$. (Otherwise, by Theorem 3.1, $m_1 = m_2$ and, therefore, $m_1^2 = -1$, which contradicts the fact that the square of a real number is never negative.) We must show that $\mathcal{L}_1$ and $\mathcal{L}_2$ are perpendicular. Let P be the intersection of $\mathcal{L}_1$ and $\mathcal{L}_2$ (see Fig. 3-14). Let $\mathcal{L}_3$ be the line through P that is perpendicular to $\mathcal{L}_1$. If m_3 is the slope of $\mathcal{L}_3$, then, by the first part of the proof, $m_1 m_3 = -1$ and, therefore, $m_1 m_3 = m_1 m_2$. Since $m_1 m_3 = -1$, $m_1 \neq 0$; therefore, $m_3 = m_2$. Since $\mathcal{L}_2$ and $\mathcal{L}_3$ pass through the same point P and have the same slope, they must coincide. Since $\mathcal{L}_1$ and $\mathcal{L}_3$ are perpendicular, $\mathcal{L}_1$ and $\mathcal{L}_2$ are also perpendicular.

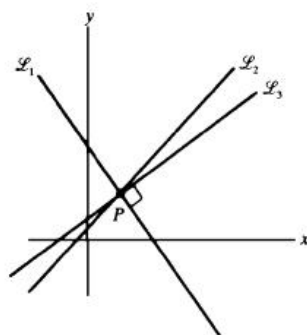


Fig. 3-14

6. Show that, if a and b are not both zero, then the equation $ax + by = c$ is the equation of a line and, conversely, every line has an equation of that form.

Assume $b \neq 0$. Then, if the equation $ax + by = c$ is solved for y , we obtain a slope-intercept equation $y = (-a/b)x + c/b$ of a line. If $b = 0$, then $a \neq 0$, and the equation $ax + by = c$ reduces to $ax = c$; this is equivalent to $x = c/a$, the equation of a vertical line.

Conversely, every nonvertical line has a slope-intercept equation $y = mx + b$, which is equivalent to $-mx + y = b$, an equation of the desired form. A vertical line has an equation of the form $x = c$, which is also an equation of the required form with $a = 1$ and $b = 0$.

7. Show that the line $y = x$ makes an angle of 45° with the positive x axis (that is, that angle BOA in Fig. 3-15 contains 45°).

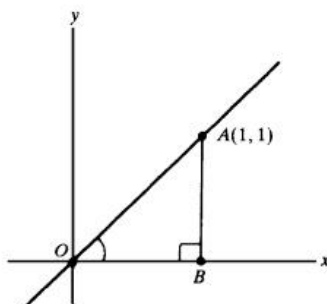


Fig. 3-15

Let A be the point on the line $y = x$ with coordinates $(1, 1)$. Drop a perpendicular AB to the positive x axis. Then $AB = 1$ and $OB = 1$. Hence, angle $OAB =$ angle BOA , since they are the base angles of isosceles triangle BOA . Since angle OBA is a right angle,

$$\text{Angle } OAB + \text{angle } BOA = 180^\circ - \text{angle } OBA = 180^\circ - 90^\circ = 90^\circ$$

Since angle $BOA =$ angle OAB , they each contain 45° .

8. Show that the distance d from a point $P(x_1, y_1)$ to a line $\mathcal{L}$ with equation $ax + by = c$ is given by the formula

$$d = \frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}}.$$

Let $\mathcal{M}$ be the line through P that is perpendicular to $\mathcal{L}$. Then $\mathcal{M}$ intersects $\mathcal{L}$ at some point Q with coordinates (u, v) , as in Fig. 3-16. Clearly, d is the length PQ , so if we can find u and v , we can compute d with the distance formula. The slope of $\mathcal{L}$ is $-a/b$. Hence, by Theorem 3.2, the slope of $\mathcal{M}$ is b/a . Then a point-slope equation of $\mathcal{M}$ is $\frac{y - y_1}{x - x_1} = \frac{b}{a}$. Thus, u and v are the solutions of the pair of equations $au + bv = c$ and $\frac{v - y_1}{u - x_1} = \frac{b}{a}$. Tedious algebraic calculations yield the solution

$$u = \frac{ac + b^2x_1 + aby_1}{a^2 + b^2} \quad \text{and} \quad v = \frac{bc - abx_1 + a^2y_1}{a^2 + b^2}$$

The distance formula, together with further calculations, yields

$$d = \overline{PQ} = \sqrt{(x_1 - u)^2 + (y_1 - v)^2} = \frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}}.$$

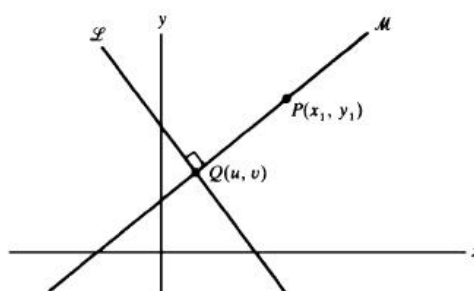


Fig. 3-16

SUPPLEMENTARY PROBLEMS

9. Find a point-slope equation for the line through each of the following pairs of points: (a) (3, 6) and (2, -4); (b) (8, 5) and (4, 0); (c) (1, 3) and the origin; (d) (2, 4) and (-2, 4).

Ans. (a) $\frac{y-6}{x-3} = 10$; (b) $\frac{y-5}{x-8} = \frac{5}{4}$; (c) $\frac{y-3}{x-1} = 3$; (d) $\frac{y-4}{x-2} = 0$

10. Find the slope-intercept equation of each line:

- Through the points (4, -2) and (1, 7)
- Having slope 3 and y intercept 4
- Through the points (-1, 0) and (0, 3)
- Through (2, -3) and parallel to the x axis
- Through (2, 3) and rising 4 units for every unit increase in x
- Through (-2, 2) and falling 2 units for every unit increase in x
- Through (3, -4) and parallel to the line with equation $5x - 2y = 4$
- Through the origin and parallel to the line with equation $y = 2$
- Through (-2, 5) and perpendicular to the line with equation $4x + 8y = 3$
- Through the origin and perpendicular to the line with equation $3x - 2y = 1$
- Through (2, 1) and perpendicular to the line with equation $x = 2$
- Through the origin and bisecting the angle between the positive x axis and the positive y axis

Ans. (a) $y = -3x + 10$; (b) $y = 3x + 3$; (c) $y = 3x + 3$; (d) $y = -3$; (e) $y = 4x - 5$; (f) $y = -2x - 2$; (g) $y = \frac{5}{2}x - \frac{23}{2}$; (h) $y = 0$; (i) $y = 2x + 9$; (j) $y = -\frac{2}{5}x$; (k) $y = 1$; (l) $y = x$

CHAPTER 3 Lines

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11. (a) Describe the lines having equations of the form $x = a$.
 (b) Describe the lines having equations of the form $y = b$.
 (c) Describe the line having the equation $y = -x$.
12. (a) Find the slopes and y intercepts of the lines that have the following equations: (i) $y = 3x - 2$; (ii) $2x - 5y = 3$; (iii) $y = 4x - 3$; (iv) $y = -3$; (v) $\frac{y}{2} + \frac{x}{3} = 1$.
 (b) Find the coordinates of a point other than $(0, b)$ on each of the lines of part (a).
- Ans. (a) (i) $m = 3, b = -2$; (ii) $m = \frac{2}{5}, b = -\frac{3}{5}$; (iii) $m = 4, b = -3$; (iv) $m = 0, b = -3$; (v) $m = -\frac{2}{3}, b = 2$;
 (b) (i) $(1, 1)$; (ii) $(-6, -3)$; (iii) $(1, 1)$; (iv) $(1, -3)$; (v) $(3, 0)$

13. If the point $(3, k)$ lies on the line with slope $m = -2$ passing through the point $(2, 5)$, find k .

Ans. $k = 3$

14. Does the point $(3, -2)$ lie on the line through the points $(8, 0)$ and $(-7, -6)$?

Ans. Yes

15. Use slopes to determine whether the points $(7, -1)$, $(10, 1)$, and $(6, 7)$ are the vertices of a right triangle.

Ans. They are.

16. Use slopes to determine whether $(8, 0)$, $(-1, -2)$, $(-2, 3)$, and $(7, 5)$ are the vertices of a parallelogram.

Ans. They are.

17. Under what conditions are the points $(u, v + w)$, $(v, u + w)$, and $(w, u + v)$ collinear?

Ans. Always.

18. Determine k so that the points $A(7, 3)$, $B(-1, 0)$, and $C(k, -2)$ are the vertices of a right triangle with right angle at B .

Ans. $k = 1$

19. Determine whether the following pairs of lines are parallel, perpendicular, or neither:

- (a) $y = 3x + 2$ and $y = 3x - 2$
 (b) $y = 2x - 4$ and $y = 3x + 5$
 (c) $3x - 2y = 5$ and $2x + 3y = 4$
 (d) $6x + 3y = 1$ and $4x + 2y = 3$
 (e) $x = 3$ and $y = -4$
 (f) $5x + 4y = 1$ and $4x + 5y = 2$
 (g) $x = -2$ and $x = 7$

Ans. (a) Parallel; (b) neither; (c) perpendicular; (d) parallel; (e) perpendicular; (f) neither; (g) parallel

20. Draw the line determined by the equation $2x + 5y = 10$. Determine whether the points $(10, 2)$ and $(12, 3)$ lie on this line.

21. For what values of k will the line $kx - 3y = 4k$ have the following properties: (a) have slope 1; (b) have y intercept 2; (c) pass through the point $(2, 4)$; (d) be parallel to the line $2x - 4y = 1$; (e) be perpendicular to the line $x - 6y = 2$?

Ans. (a) $k = 3$; (b) $k = -\frac{3}{2}$; (c) $k = -6$; (d) $k = \frac{3}{2}$; (e) $k = -18$

22. Describe geometrically the families of lines (a) $y = mx - 3$ and (b) $y = 4x + b$, where m and b are any real numbers.

Ans. (a) Lines with y intercept -3 ; (b) lines with slope 4

23. In the triangle with vertices, $A(0, 0)$, $B(2, 0)$, and $C(3, 3)$, find equations for (a) the median from B to the midpoint of the opposite side; (b) the perpendicular bisector of side BC ; and (c) the altitude from B to the opposite side.

Ans. (a) $y = -3x + 6$; (b) $x + 3y = 7$; (c) $y = -x + 2$

CHAPTER 4

Circles

Equations of Circles

For a point $P(x, y)$ to lie on the circle with center $C(a, b)$ and radius r , the distance $\overline{PC}$ must be equal to r (see Fig. 4-1). By the distance formula (2.1),

$$\overline{PC} = \sqrt{(x-a)^2 + (y-b)^2}$$

Thus, P lies on the circle if and only if

$$(x-a)^2 + (y-b)^2 = r^2 \quad (4.1)$$

Equation (4.1) is called the *standard equation* of the circle with center at (a, b) and radius r .

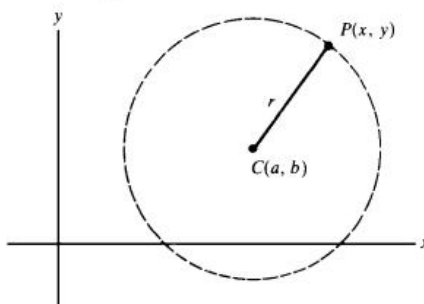


Fig. 4-1

EXAMPLE 4.1:

- (a) The circle with center $(3, 1)$ and radius 2 has the equation $(x-3)^2 + (y-1)^2 = 4$.
- (b) The circle with center $(2, -1)$ and radius 3 has the equation $(x-2)^2 + (y+1)^2 = 9$.
- (c) What is the set of points satisfying the equation $(x-4)^2 + (y-5)^2 = 25$?

By (4.1), this is the equation of the circle with center at $(4, 5)$ and radius 5. That circle is said to be the *graph* of the given equation, that is, the set of points satisfying the equation.

- (d) The graph of the equation $(x+3)^2 + y^2 = 2$ is the circle with center at $(-3, 0)$ and radius $\sqrt{2}$.

The Standard Equation of a Circle

The standard equation of a circle with center at the origin $(0, 0)$ and radius r is

$$x^2 + y^2 = r^2 \quad (4.2)$$

For example, $x^2 + y^2 = 1$ is the equation of the circle with center at the origin and radius 1. The graph of $x^2 + y^2 = 5$ is the circle with center at the origin and radius $\sqrt{5}$.

The equation of a circle sometimes appears in a disguised form. For example, the equation

$$x^2 + y^2 + 8x - 6y + 21 = 0 \quad (4.3)$$

turns out to be equivalent to

$$(x+4)^2 + (y-3)^2 = 4 \quad (4.4)$$

Equation (4.4) is the standard equation of a circle with center at $(-4, 3)$ and radius 2.

Equation (4.4) is obtained from (4.3) by a process called *completing the square*. In general terms, the process involves finding the number that must be added to the sum $x^2 + Ax$ to obtain a square.

Here, we note that $\left(x + \frac{A}{2}\right)^2 = x^2 + Ax + \left(\frac{A}{2}\right)^2$. Thus, in general, we *must add* $\left(\frac{A}{2}\right)^2$ to $x^2 + Ax$ to obtain the square $\left(x + \frac{A}{2}\right)^2$. For example, to get a square from $x^2 + 8x$, we add $\left(\frac{8}{2}\right)^2$, that is, 16.

The result is $x^2 + 8x + 16$, which is $(x+4)^2$. This is the process of completing the square.

Consider the original (4.3): $x^2 + y^2 + 8x - 6y + 21 = 0$. To complete the square in $x^2 + 8x$, we add 16. To complete the square in $y^2 - 6y$, we add $\left(-\frac{6}{2}\right)^2$, which is 9. Of course, since we added 16 and 9 to the left side of the equation, we must also add them to the right side, obtaining

$$(x^2 + 8x + 16) + (y^2 - 6y + 9) + 21 = 16 + 9$$

This is equivalent to

$$(x+4)^2 + (y-3)^2 + 21 = 25$$

and subtraction of 21 from both sides yields (4.4).

EXAMPLE 4.2: Consider the equation $x^2 + y^2 - 4x - 10y + 20 = 0$. Completing the square yields

$$\begin{aligned} (x^2 - 4x + 4) + (y^2 - 10y + 25) + 20 &= 4 + 25 \\ (x-2)^2 + (y-5)^2 &= 9 \end{aligned}$$

Thus, the original equation is the equation of a circle with center at $(2, 5)$ and radius 3.

The process of completing the square can be applied to any equation of the form

$$x^2 + y^2 + Ax + By + C = 0 \quad (4.5)$$

to obtain

$$\left(x + \frac{A}{2}\right)^2 + \left(y + \frac{B}{2}\right)^2 + C = \frac{A^2}{4} + \frac{B^2}{4}$$

or

$$\left(x + \frac{A}{2}\right)^2 + \left(y + \frac{B}{2}\right)^2 = \frac{A^2 + B^2 - 4C}{4} \quad (4.6)$$

There are three different cases, depending on whether $A^2 + B^2 - 4C$ is positive, zero, or negative.

Case 1: $A^2 + B^2 - 4C > 0$. In this case, (4.6) is the standard equation of a circle with center at $\left(-\frac{A}{2}, -\frac{B}{2}\right)$ and radius $\frac{\sqrt{A^2 + B^2 - 4C}}{2}$.

Case 2: $A^2 + B^2 - 4C = 0$. A sum of the squares of two quantities is zero when and only when each of the quantities is zero. Hence, (4.6) is equivalent to the conjunction of the equations $x + A/2 = 0$ and $y + B/2 = 0$ in this case, and the only solution of (4.6) is the point $(-A/2, -B/2)$. Hence, the graph of (4.5) is a single point, which may be considered a *degenerate circle* of radius 0.

Case 3: $A^2 + B^2 - 4C < 0$. A sum of two squares cannot be negative. So, in this case, (4.5) has no solution at all.

We can show that any circle has an equation of the form (4.5). Suppose its center is (a, b) and its radius is r ; then its standard equation is

$$(x-a)^2 + (y-b)^2 = r^2$$

Expanding yields $x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2$, or

$$x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) = 0$$

CHAPTER 4 Circles

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SOLVED PROBLEMS

1. Identify the graphs of (a) $2x^2 + 2y^2 - 4x + y + 1 = 0$; (b) $x^2 + y^2 - 4y + 7 = 0$; (c) $x^2 + y^2 - 6x - 2y + 10 = 0$.

(a) First divide by 2, obtaining $x^2 + y^2 - 2x + \frac{1}{2}y + \frac{1}{2} = 0$. Then complete the squares:

$$(x^2 - 2x + 1) + (y^2 + \frac{1}{2}y + \frac{1}{16}) + \frac{1}{2} = 1 + \frac{1}{16} = \frac{17}{16}$$

$$(x - 1)^2 + (y + \frac{1}{4})^2 = \frac{17}{16} - \frac{1}{2} = \frac{17}{16} - \frac{8}{16} = \frac{9}{16}$$

Thus, the graph is the circle with center $(1, -\frac{1}{4})$ and radius $\frac{3}{4}$.

(b) Complete the square:

$$x^2 + (y - 2)^2 + 7 = 4$$

$$x^2 + (y - 2)^2 = -3$$

Because the right side is negative, there are no points in the graph.

(c) Complete the square:

$$(x - 3)^2 + (y - 1)^2 + 10 = 9 + 1$$

$$(x - 3)^2 + (y - 1)^2 = 0$$

The only solution is the point $(3, 1)$.

2. Find the standard equation of the circle with center at $C(2, 3)$ and passing through the point $P(-1, 5)$.

The radius of the circle is the distance

$$\overline{CP} = \sqrt{(5 - 3)^2 + (-1 - 2)^2} = \sqrt{2^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$$

So the standard equation is $(x - 2)^2 + (y - 3)^2 = 13$.

3. Find the standard equation of the circle passing through the points $P(3, 8)$, $Q(9, 6)$, and $R(13, -2)$.

First method: The circle has an equation of the form $x^2 + y^2 + Ax + By + C = 0$. Substitute the values of x and y at point P , to obtain $9 + 64 + 3A + 8B + C = 0$ or

$$3A + 8B + C = -73 \quad (1)$$

A similar procedure for points Q and R yields the equations

$$9A + 6B + C = -117 \quad (2)$$

$$13A - 2B + C = -173 \quad (3)$$

Eliminate C from (1) and (2) by subtracting (2) from (1):

$$-6A + 2B = 44 \quad \text{or} \quad -3A + B = 22 \quad (4)$$

Eliminate C from (1) and (3) by subtracting (3) from (1):

$$-10A + 10B = 100 \quad \text{or} \quad -A + B = 10 \quad (5)$$

Eliminate B from (4) and (5) by subtracting (5) from (4), obtaining $A = -6$. Substitute this value in (5) to find that $B = 4$. Then solve for C in (1): $C = -87$.

Hence, the original equation for the circle is $x^2 + y^2 - 6x + 4y - 87 = 0$. Completing the squares then yields

$$(x - 3)^2 + (y + 2)^2 = 87 + 9 + 4 = 100$$

Thus, the circle has center $(3, -2)$ and radius 10.

Second method: The perpendicular bisector of any chord of a circle passes through the center of the circle. Hence, the perpendicular bisector $\mathcal{L}$ of chord PQ will intersect the perpendicular bisector $\mathcal{M}$ of chord QR at the center of the circle (see Fig. 4-2).

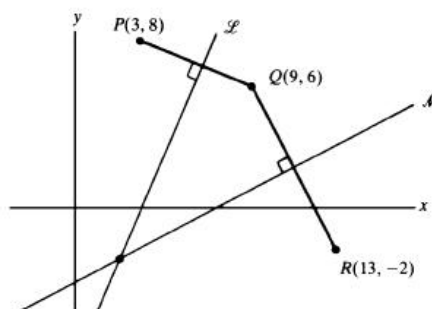


Fig. 4-2

The slope of line PQ is $-\frac{1}{3}$. So, by Theorem 3.2, the slope of $\mathcal{L}$ is 3. Also, $\mathcal{L}$ passes through the midpoint $(6, 7)$ of segment PQ . Hence a point-slope equation of $\mathcal{L}$ is $\frac{y-7}{x-6} = 3$, and therefore its slope-intercept equation is $y = 3x - 11$. Similarly, the slope of line QR is -2 , and therefore the slope of $\mathcal{M}$ is $\frac{1}{2}$. Since $\mathcal{M}$ passes through the midpoint $(11, 2)$ of segment QR , it has a point-slope equation $\frac{y-2}{x-11} = \frac{1}{2}$, which yields the slope-intercept equation $y = \frac{1}{2}x - \frac{7}{2}$. Hence, the coordinates of the center of the circle satisfy the two equations $y = 3x - 11$ and $y = \frac{1}{2}x - \frac{7}{2}$ and we may write

$$3x - 11 = \frac{1}{2}x - \frac{7}{2}$$

from which we find that $x = 3$. Therefore,

$$y = 3x - 11 = 3(3) - 11 = -2$$

So the center is at $(3, -2)$. The radius is the distance between the center and the point $(3, 8)$:

$$\sqrt{(-2-8)^2 + (3-3)^2} = \sqrt{(-10)^2} = \sqrt{100} = 10$$

Thus, the standard equation of the circle is $(x-3)^2 + (y+2)^2 = 100$.

4. Find the center and radius of the circle that passes through $P(1, 1)$ and is tangent to the line $y = 2x - 3$ at the point $Q(3, 3)$. (See Fig. 4-3.)

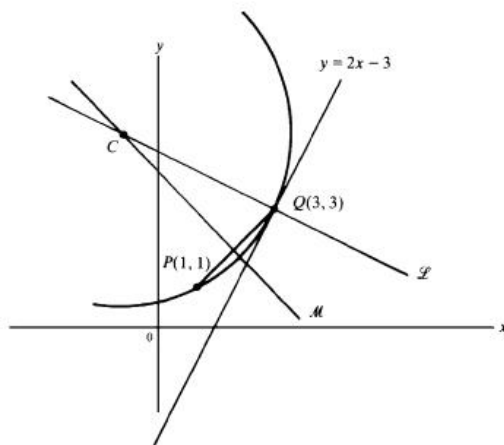


Fig. 4-3

The line $\mathcal{L}$ perpendicular to $y = 2x - 3$ at $(3, 3)$ must pass through the center of the circle. By Theorem 3.2, the slope of $\mathcal{L}$ is $-\frac{1}{2}$. Therefore, the slope-intercept equation of $\mathcal{L}$ has the form $y = -\frac{1}{2}x + b$. Since $(3, 3)$ is on $\mathcal{L}$, we have $3 = -\frac{1}{2}(3) + b$; hence, $b = \frac{9}{2}$, and $\mathcal{L}$ has the equation $y = -\frac{1}{2}x + \frac{9}{2}$.

CHAPTER 4 Circles

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The perpendicular bisector $\mathcal{M}$ of chord PQ in Fig. 4-3 also passes through the center of the circle, so the intersection of $\mathcal{L}$ and $\mathcal{M}$ will be the center of the circle. The slope of PQ is 1. Hence, by Theorem 3.2, the slope of $\mathcal{M}$ is -1 . So $\mathcal{M}$ has the slope-intercept equation $y = -x + b'$. Since the midpoint $(2, 2)$ of chord PQ is a point on $\mathcal{M}$, we have $2 = -(2) + b'$; hence, $b' = 4$, and the equation of $\mathcal{M}$ is $y = -x + 4$. We must find the common solution of $y = -x + 4$ and $y = -\frac{1}{2}x + \frac{9}{2}$. Setting

$$-x + 4 = -\frac{1}{2}x + \frac{9}{2}$$

yields $x = -1$. Therefore, $y = -x + 4 = -(-1) + 4 = 5$, and the center C of the circle is $(-1, 5)$. The radius is the distance $\overline{PC} = \sqrt{(-1-3)^2 + (5-3)^2} = \sqrt{16+4} = \sqrt{20}$. The standard equation of the circle is then $(x+1)^2 + (y-5)^2 = 20$.

5. Find the standard equation of every circle that passes through the points $P(1, -1)$ and $Q(3, 1)$ and is tangent to the line $y = -3x$.

Let $C(c, d)$ be the center of one of the circles, and let A be the point of tangency (see Fig. 4-4). Then, because $\overline{CP} = \overline{CQ}$, we have

$$\overline{CP}^2 = \overline{CQ}^2 \quad \text{or} \quad (c-1)^2 + (d+1)^2 = (c-3)^2 + (d-1)^2$$

Expanding and simplifying, we obtain

$$c + d = 2 \tag{1}$$

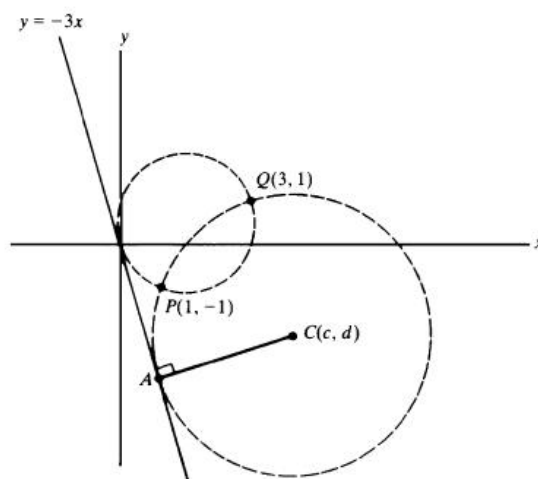


Fig. 4-4

In addition, $\overline{CP} = \overline{CA}$, and by the formula of Problem 8 in Chapter 3, $\overline{CA} = \frac{3c+d}{\sqrt{10}}$. Setting $\overline{CP}^2 = \overline{CA}^2$ thus yields $(c-1)^2 + (d+1)^2 = \frac{(3c+d)^2}{10}$. Substituting (1) in the right-hand side and multiplying by 10 then yields

$$10[(c-1)^2 + (d+1)^2] = (2c+2)^2 \quad \text{from which} \quad 3c^2 + 5d^2 - 14c + 10d + 8 = 0$$

By (1), we can replace d by $2 - c$, obtaining

$$2c^2 - 11c + 12 = 0 \quad \text{or} \quad (2c-3)(c-4) = 0$$

Hence, $c = \frac{3}{2}$ or $c = 4$. Then (1) gives us the two solutions $c = \frac{3}{2}, d = \frac{1}{2}$ and $c = 4, d = -2$. Since the radius $\overline{CA} = \frac{3c+d}{\sqrt{10}}$, these solutions produce radii of $\frac{\frac{10}{2}}{\sqrt{10}} = \frac{\sqrt{10}}{2}$ and $\frac{10}{\sqrt{10}} = \sqrt{10}$. Thus, there are two such circles, and

their standard equations are

$$\left(x - \frac{3}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{5}{2} \quad \text{and} \quad (x - 4)^2 + (y + 2)^2 = 10$$

SUPPLEMENTARY PROBLEMS

6. Find the standard equations of the circles satisfying the following conditions:

- (a) center at (3, 5) and radius 2 (b) center at (4, -1) and radius 1
 (c) center at (5, 0) and radius $\sqrt{3}$ (d) center at (-2, -2) and radius $5\sqrt{2}$
 (e) center at (-2, 3) and passing through (3, -2) (f) center at (6, 1) and passing through the origin

Ans. (a) $(x - 3)^2 + (y - 5)^2 = 4$; (b) $(x - 4)^2 + (y + 1)^2 = 1$; (c) $(x - 5)^2 + y^2 = 3$; (d) $(x + 2)^2 + (y + 2)^2 = 50$;
 (e) $(x + 2)^2 + (y - 3)^2 = 50$; (f) $(x - 6)^2 + (y - 1)^2 = 37$

7. Identify the graphs of the following equations:

- (a) $x^2 + y^2 + 16x - 12y + 10 = 0$ (b) $x^2 + y^2 - 4x + 5y + 10 = 0$
 (c) $x^2 + y^2 + x - y = 0$ (d) $4x^2 + 4y^2 + 8y - 3 = 0$
 (e) $x^2 + y^2 - x - 2y + 3 = 0$ (f) $x^2 + y^2 + \sqrt{2}x - 2 = 0$

Ans. (a) circle, center at (-8, 6), radius $3\sqrt{10}$; (b) circle, center at $(2, -\frac{5}{2})$, radius $\frac{1}{2}$; (c) circle, center at $(-\frac{1}{2}, \frac{1}{2})$, radius $\frac{\sqrt{2}}{2}$; (d) circle, center at (0, -1), radius $\frac{3}{2}$; (e) empty graph; (f) circle, center at $(-\sqrt{2}/2, 0)$, radius $\sqrt{5/2}$

8. Find the standard equations of the circles through (a) (-2, 1), (1, 4), and (-3, 2); (b) (0, 1), (2, 3), and $(1, 1 + \sqrt{3})$; (c) (6, 1), (2, -5), and (1, -4); (d) (2, 3), (-6, -3), and (1, 4).

Ans. (a) $(x + 1)^2 + (y - 3)^2 = 5$; (b) $(x - 2)^2 + (y - 1)^2 = 4$; (c) $(x - 4)^2 + (y + 2)^2 = 13$; (d) $(x + 2)^2 + y^2 = 25$

9. For what values of k does the circle $(x + 2k)^2 + (y - 3k)^2 = 10$ pass through the point (1, 0)?

Ans. $k = \frac{9}{13}$ or $k = -1$

10. Find the standard equations of the circles of radius 2 that are tangent to both the lines $x = 1$ and $y = 3$.

Ans. $(x + 1)^2 + (y - 1)^2 = 4$; $(x + 1)^2 + (y - 5)^2 = 4$; $(x - 3)^2 + (y - 1)^2 = 4$; $(x - 3)^2 + (y - 5)^2 = 4$

11. Find the value of k so that $x^2 + y^2 + 4x - 6y + k = 0$ is the equation of a circle of radius 5.

Ans. $k = -12$

12. Find the standard equation of the circle having as a diameter the segment joining (2, -3) and (6, 5).

Ans. $(x - 4)^2 + (y - 1)^2 = 20$

13. Find the standard equation of every circle that passes through the origin, has radius 5, and is such that the y coordinate of its center is -4.

Ans. $(x - 3)^2 + (y + 4)^2 = 25$ or $(x + 3)^2 + (y + 4)^2 = 25$

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14. Find the standard equation of the circle that passes through the points $(8, -5)$ and $(-1, 4)$ and has its center on the line $2x + 3y = 3$.

Ans. $(x - 3)^2 + (y + 1)^2 = 41$

15. Find the standard equation of the circle with center $(3, 5)$ that is tangent to the line $12x - 5y + 2 = 0$.

Ans. $(x - 3)^2 + (y - 5)^2 = 1$

16. Find the standard equation of the circle that passes through the point $(1, 3 + \sqrt{2})$ and is tangent to the line $x + y = 2$ at $(2, 0)$.

Ans. $(x - 5)^2 + (y - 3)^2 = 18$

17. Prove analytically that an angle inscribed in a semicircle is a right angle. (See Fig. 4-5.)

18. Find the length of a tangent from $(6, -2)$ to the circle $(x - 1)^2 + (y - 3)^2 = 1$. (See Fig. 4-6.)

Ans. 7

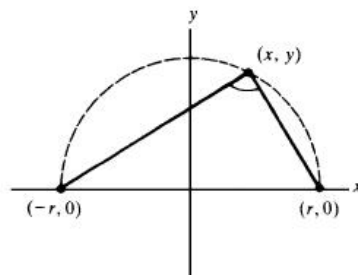


Fig. 4-5

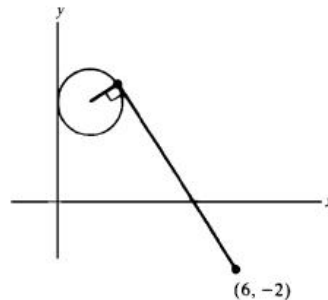


Fig. 4-6

19. Find the standard equations of the circles that pass through $(2, 3)$ and are tangent to both the lines $3x - 4y = -1$ and $4x + 3y = 7$.

Ans. $(x - 2)^2 + (y - 8)^2 = 25$ and $\left(x - \frac{6}{5}\right)^2 + \left(y - \frac{12}{5}\right)^2 = 1$

20. Find the standard equations of the circles that have their centers on the line $4x + 3y = 8$ and are tangent to both the lines $x + y = -2$ and $7x - y = -6$.

Ans. $(x - 1)^2 + y^2 = 2$ and $(x + 4)^2 + (y - 8)^2 = 18$

21. Find the standard equation of the circle that is concentric with the circle $x^2 + y^2 - 2x - 8y + 1 = 0$ and is tangent to the line $2x - y = 3$.

Ans. $(x - 1)^2 + (y - 4)^2 = 5$

22. Find the standard equations of the circles that have radius 10 and are tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Ans. $(x - 9)^2 + (y - 12)^2 = 100$ and $(x + 3)^2 + (y + 4)^2 = 100$

23. Find the longest and shortest distances from the point (7, 12) to the circle $x^2 + y^2 + 2x + 6y - 15 = 0$.

Ans. 22 and 12

24. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be two intersecting circles determined by the equations $x^2 + y^2 + A_1x + B_1y + C_1 = 0$ and $x^2 + y^2 + A_2x + B_2y + C_2 = 0$. For any number $k \neq -1$, show that

$$x^2 + y^2 + A_1x + B_1y + C_1 + k(x^2 + y^2 + A_2x + B_2y + C_2) = 0$$

is the equation of a circle through the intersection points of $\mathcal{C}_1$ and $\mathcal{C}_2$. Show, conversely, that every such circle may be represented by such an equation for a suitable k .

25. Find the standard equation of the circle passing through the point $(-3, 1)$ and containing the points of intersection of the circles $x^2 + y^2 + 5x = 1$ and $x^2 + y^2 + y = 7$.

Ans. (Use Problem 24.) $(x+1)^2 + \left(y + \frac{3}{10}\right)^2 = \frac{569}{100}$

26. Find the standard equations of the circles that have centers on the line $5x - 2y = -21$ and are tangent to both coordinate axes.

Ans. $(x+7)^2 + (y+7)^2 = 49$ and $(x+3)^2 + (y-3)^2 = 9$

27. (a) If two circles $x^2 + y^2 + A_1x + B_1y + C_1 = 0$ and $x^2 + y^2 + A_2x + B_2y + C_2 = 0$ intersect at two points, find an equation of the line through their points of intersection.

(b) Prove that if two circles intersect at two points, then the line through their points of intersection is perpendicular to the line through their centers.

Ans. (a) $(A_1 - A_2)x + (B_1 - B_2)y + (C_1 - C_2) = 0$

28. Find the points of intersection of the circles $x^2 + y^2 + 8y - 64 = 0$ and $x^2 + y^2 - 6x - 16 = 0$.

Ans. $(8, 0)$ and $\left(\frac{24}{5}, \frac{24}{5}\right)$

29. Find the equations of the lines through $(4, 10)$ and tangent to the circle $x^2 + y^2 - 4y - 36 = 0$.

Ans. $y = -3x + 22$ and $x - 3y + 26 = 0$

30. (GC) Use a graphing calculator to draw the circles in Problems 7(d), 10, 14, and 15. (Note: It may be necessary to solve for y .)

31. (GC) (a) Use a graphing calculator to shade the interior of the circle with center at the origin and radius 3.

(b) Use a graphing calculator to shade the exterior of the circle $x^2 + (y-2)^2 = 1$.

32. (GC) Use a graphing calculator to graph the following inequalities: (a) $(x-1)^2 + y^2 < 4$; (b) $x^2 + y^2 - 6x - 8y > 0$.

CHAPTER 5

Equations and Their Graphs

The Graph of an Equation

The graph of an equation involving x and y as its only variables consists of all points (x, y) satisfying the equation.

EXAMPLE 5.1: (a) What is the graph of the equation $2x - y = 3$?

The equation is equivalent to $y = 2x - 3$, which we know is the slope-intercept equation of the line with slope 2 and y intercept -3 .

(b) What is the graph of the equation $x^2 + y^2 - 2x + 4y - 4 = 0$?

Completing the square shows that the given equation is equivalent to the equation $(x - 1)^2 + (y + 2)^2 = 9$. Hence, its graph is the circle with center $(1, -2)$ and radius 3.

Parabolas

Consider the equation $y = x^2$. If we substitute a few values for x and calculate the associated values of y , we obtain the results tabulated in Fig. 5-1. We can plot the corresponding points, as shown in the figure. These points suggest the heavy curve, which belongs to a family of curves called *parabolas*. In particular, the graphs of equations of the form $y = cx^2$, where c is a nonzero constant, are parabolas, as are any other curves obtained from them by translations and rotations.

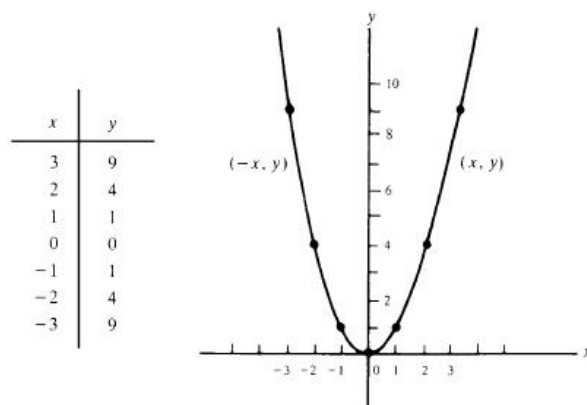


Fig. 5-1

In Fig. 5-1, we note that the graph of $y = x^2$ contains the origin $(0, 0)$ but all its other points lie above the x axis, since x^2 is positive except when $x = 0$. When x is positive and increasing, y increases without bound. Hence, in the first quadrant, the graph moves up without bound as it moves right. Since $(-x)^2 = x^2$, it follows that, if any point (x, y) lies on the graph in the first quadrant, then the point $(-x, y)$ also lies on the graph in the second quadrant. Thus, the graph is symmetric with respect to the y axis. The y axis is called the *axis of symmetry* of this parabola.

Ellipses

To construct the graph of the equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$, we again compute a few values and plot the corresponding points, as shown in Fig. 5-2. The graph suggested by these points is also drawn in the figure; it is a member of a family of curves called *ellipses*. In particular, the graph of an equation of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is an ellipse, as is any curve obtained from it by translation or rotation.

Note that, in contrast to parabolas, ellipses are bounded. In fact, if (x, y) is on the graph of $\frac{x^2}{9} + \frac{y^2}{4} = 1$, then $\frac{x^2}{9} \leq \frac{x^2}{9} + \frac{y^2}{4} = 1$, and, therefore, $x^2 \leq 9$. Hence, $-3 \leq x \leq 3$. So, the graph lies between the vertical lines $x = -3$ and $x = 3$. Its rightmost point is $(3, 0)$, and its leftmost point is $(-3, 0)$. A similar argument shows that the graph lies between the horizontal lines $y = -2$ and $y = 2$, and that its lowest point is $(0, -2)$ and its highest point is $(0, 2)$. In the first quadrant, as x increases from 0 to 3, y decreases from 2 to 0. If (x, y) is any point on the graph, then $(-x, y)$ also is on the graph. Hence, the graph is symmetric with respect to the y axis. Similarly, if (x, y) is on the graph, so is $(x, -y)$, and therefore the graph is symmetric with respect to the x axis.

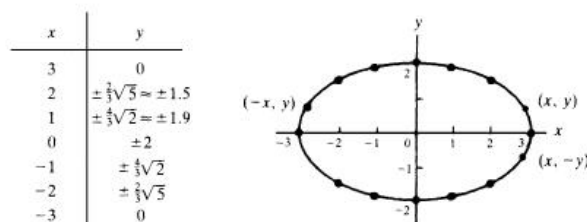


Fig. 5-2

When $a = b$, the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is the circle with the equation $x^2 + y^2 = a^2$, that is, a circle with center at the origin and radius a . Thus, circles are special cases of ellipses.

Hyperbolas

Consider the graph of the equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$. Some of the points on this graph are tabulated and plotted in Fig. 5-3. These points suggest the curve shown in the figure, which is a member of a family of curves called *hyperbolas*. The graphs of equations of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are hyperbolas, as are any curves obtained from them by translations and rotations.

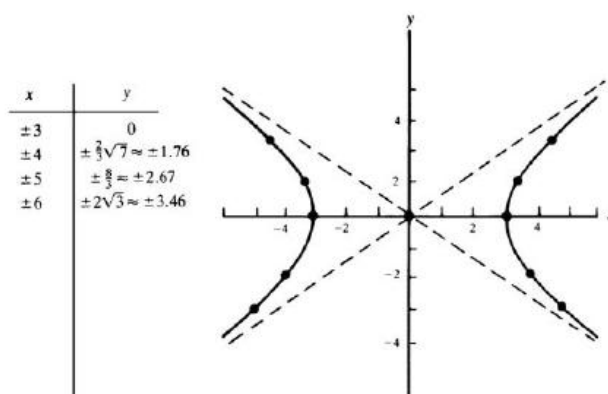


Fig. 5-3

Let us look at the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ in more detail. Since $\frac{x^2}{9} = 1 + \frac{y^2}{4} \geq 1$, it follows that $x^2 \geq 9$, and therefore, $|x| \geq 3$. Hence, there are no points on the graph between the vertical lines $x = -3$ and $x = 3$. If (x, y)

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is on the graph, so is $(-x, y)$; thus, the graph is symmetric with respect to the y axis. Similarly, the graph is symmetric with respect to the x axis. In the first quadrant, as x increases, y increases without bound.

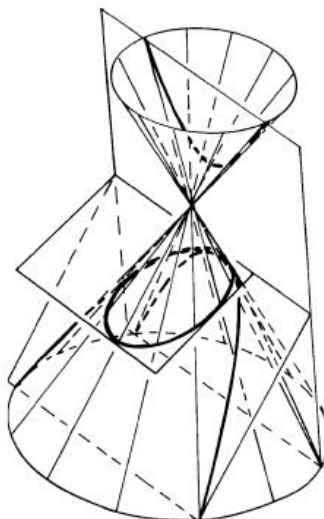


Fig. 5-4

Note the dashed lines in Fig. 5-3; they are the lines $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$, and they are called the *asymptotes* of the hyperbola: Points on the hyperbola get closer and closer to these asymptotes as they recede from the origin. In general, the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are the lines $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$.

Conic Sections

Parabolas, ellipses, and hyperbolas together make up a class of curves called *conic sections*. They can be defined geometrically as the intersections of planes with the surface of a right circular cone, as shown in Fig. 5-4.

SOLVED PROBLEMS

1. Sketch the graph of the *cubic curve* $y = x^3$.

The graph passes through the origin $(0, 0)$. Also, for any point (x, y) on the graph, x and y have the same sign; hence, the graph lies in the first and third quadrants. In the first quadrant, as x increases, y increases without bound. Moreover, if (x, y) lies on the graph, then $(-x, -y)$ also lies on the graph. Since the origin is the midpoint of the segment connecting the points (x, y) and $(-x, -y)$, the graph is symmetric with respect to the origin. Some points on the graph are tabulated and shown in Fig. 5-5; these points suggest the heavy curve in the figure.

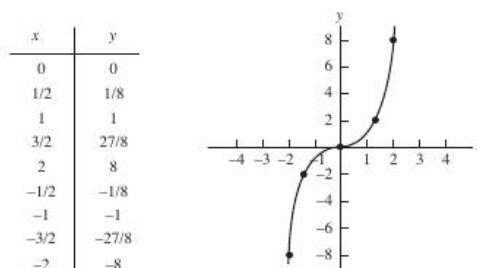


Fig. 5-5

2. Sketch the graph of the equation $y = -x^2$.

If (x, y) is on the graph of the parabola $y = x^2$ (Fig. 5-1), then $(x, -y)$ is on the graph of $y = -x^2$, and vice versa. Hence, the graph of $y = -x^2$ is the reflection in the x axis of the graph of $y = x^2$. The result is the parabola in Fig. 5-6.

3. Sketch the graph of $x = y^2$.

This graph is obtained from the parabola $y = x^2$ by exchanging the roles of x and y . The resulting curve is a parabola with the x axis as its axis of symmetry and its “nose” at the origin (see Fig. 5-7). A point (x, y) is on the graph of $x = y^2$ if and only if (y, x) is on the graph of $y = x^2$. Since the segment connecting the points (x, y) and (y, x) is perpendicular to the diagonal line $y = x$ (why?), and the midpoint $\left(\frac{x+y}{2}, \frac{x+y}{2}\right)$ of that segment is on the line $y = x$ (see Fig. 5-8), the parabola $x = y^2$ is obtained from the parabola $y = x^2$ by reflection in the line $y = x$.

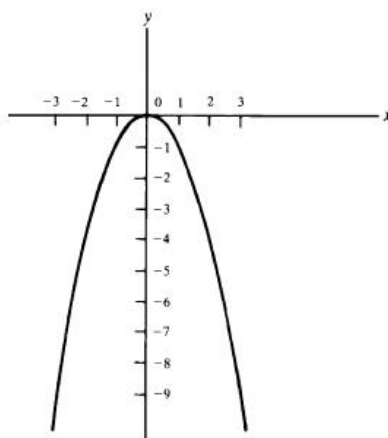


Fig. 5-6

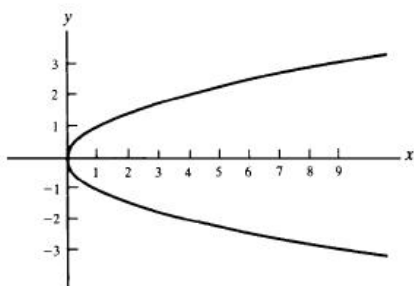


Fig. 5-7

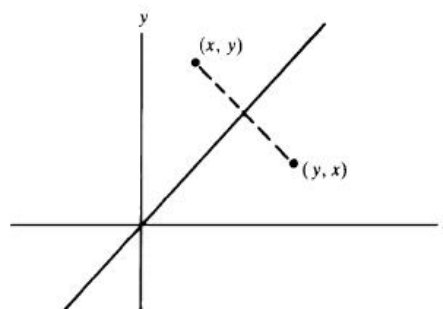


Fig. 5-8

4. Let $\mathcal{L}$ be a line, and let F be a point not on $\mathcal{L}$. Show that the set of all points equidistant from F and $\mathcal{L}$ is a parabola.

Construct a coordinate system such that F lies on the positive y axis, and the x axis is parallel to $\mathcal{L}$ and halfway between F and $\mathcal{L}$. (See Fig. 5-9.) Let $2p$ be the distance between F and $\mathcal{L}$. Then $\mathcal{L}$ has the equation $y = -p$, and the coordinates of F are $(0, p)$.